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Front cover: Pilot vessel, Chinook; tugs Bunbury and sister; fishing vessel Arctic Endurance.





REMOY VIKING: unique triple-rig shrimp trawler

Norwegian operator Jack Remoy took delivery of the world's first large triple-rig shrimp trawler, Remoy Viking, in the latter months of 2000. The vessel, whose fishing gear system has been described as unique, was built in Denmark at Orskov Christensens Skibsværft to a Norwegian design from Ulstein Nordvestconsult.

Measuring 61.00m x 14.65m, the vessel is powered by a Wärtsilä 5,520kW main engine, which is coupled to a 3,800mm diameter CP Ulstein propeller. Ulstein also supplied the rudder, steering gear and a 400kW bowthruster. The vessel has a top speed of 17 knots.

The shrimp processing plant, supplied by Danish company Carnitech, would be deemed advanced on a much bigger ship but for a vessel of this size definitely sets the trawler apart from the norm. Highlights include a new style separator which sorts the catch as it comes from the pounds and delivers clean shrimp straight into the processing line. After passing through a receiving tank, the shrimp go through three graders where they are automatically sorted by seven sizes in preparation for different processing modes. Two grades of shrimp can then be cooked simultaneously in a double batch cooker before storage in holding tanks. All shrimp then pass on to an inspection belt before being packed.

The 800m³ capacity fishroom is positioned in the centre of the vessel adjacent to the 200m³ packing hold.

The winch arrangement for Remoy Viking is largely from Rolls-Royce Ulstein Brattvaag with 'helper' winches supplied by Kinde & Toft. Nets were provided by Cosmos Trawl. The main deck has four trawl lanes extending the full length of the vessel, with eight 17-tonne sweepline winches, allowing four trawls to be ready for use at any time.

Four 45-tonne trawl winches with 47m/min handling capacity are positioned on the upper deck and controlled by means of a Synchro system from Rolls-Royce Brattvaag. Two hydraulic deck cranes - one of 6t/12m reach and another of 2t/8m reach - supplied by Dreggen Crane, are mounted to starboard and at the stern.

Wheelhouse electronics are fairly comprehensive and include a full GMDSS Satcom C installation, Navtex and extensive VHF's.

Main Particulars

Builder: Orskov Christensens Skibsværft
Vessel name: Remoy Viking
Owner/Operator: Jack Remoy, Norway
Designer: Ulstein Nordvestconsult
Total number of sister ships already completed (excluding this ship): 0

Total number of sister ships still on order: 0
Length oa: 61.00m
Length bp: 54.80m
Breadth, moulded: 14.65m
Depth, shelter deck: 8.95m
Depth, main deck: 6.15m
Service speed: 17 knots
Classification: DNV +1A1 Ice 1C with notation '1A' for the hull and Ice 1B for the rudder

Main Engine
Manufacturer: Wärtsilä
Number: 1
Output: 5,520kW

Generator
Number: 1
Make/type: Cummins
Output: 1,360kW

Propeller
Manufacturer: Ulstein Propeller
Number: 1
Fixed pitch/controllable pitch: CP
Diameter: 3,800mm

Bow Thruster
Make: Ulstein
Number: 1
Output: 400kW

Deck Machinery
Rolls-Royce Ulstein Brattvaag: 4 x trawl winches; 8 x sweepline winches; 2 x gilson winches; 2 x codend winches; 2 x overhaul winches. 2 x Aukra hydraulic deck cranes

Capacities
Fuel: 940m³
Potable water: 50 tonnes
Fishroom: 800m³
Packaging: 200m³



VILHELM THORSTEINSSON: Iceland's 'biggest' fishes for varied species

An advanced 79m x 16m stern trawler/purse seiner, *Vilhelm Thorsteinsson*, became the largest vessel in the Icelandic fleet when delivered in September 2000. Designed by Vik Sandvik and built at Kleven Verft in Norway on a hull completed at Northern Shipyard, Poland, the vessel is fitted out with the most up to date trawl gear to enable it to fish for a wide variety of species.

Able to take advantage of fish quotas that have not as yet been restricted in Icelandic waters, *Vilhelm Thorsteinsson* is equipped to catch shrimp, pelagic and bottom fish. Its Icelandic owners, Samherji hf, have installed an extensive onboard processing factory, largely aimed at increasing turn-around time for its blue whiting catches.

Hampidjan of Reykjavik, Iceland, supplied the ship with its patented Gloria 2,048m circumference self-spreading trawl, which is claimed to lower fuel costs and boost fishing performance. This is complemented by a set of Poly-Ice Super Foil mid-water trawl doors of FLS 13 type, manufactured from exceptionally strong steel but of special light construction for shallow water fishing.

Careful thought was given to the fish handling system which consists of Tendos processing lines and vacuum system; a 74 tonne/24-hour slurry ice system from Frost of Iceland and an MMC Fodema freezing arrangement. The latter includes seven MMC plate freezers with a total capacity of 100 tonnes per 24 hours.

Vilhelm Thorsteinsson is powered by a Wärtsilä 12V32 main diesel of 5,520kW. This drives a Wärtsilä CP100 propeller of 3,800mm diameter via a Wärtsilä SCV 95-P58 reduction gear. The propeller turns in an FN 3800/50 nozzle, also from Wärtsilä. It also has two Brunvoll thrusters - a 650kW FU-63-LTC-1550 unit fitted forward and an

FU-63-LTC-1750 unit of 735kW. Top speed is an impressive 18.3 knots and bollard pull, 90 tonnes.

Electrical power is generated by an ABB AMG 560 L6 shaft generator of 2,500kW, two Mitsubishi auxiliary engines (an MAS 1330 of 1,091kW and an MAS 580 of 544kW) and a Mitsubishi MAS 95 emergency generator of 84kW.

Most of the purse-trawler's navigation and fish-finding equipment is from Furuno although Simrad supplied an FS-925/903 trawl sonar system. Furuno kit includes: an FAR-2835S S-band radar; FR-2115 X-band radar; FSV-24 sonar; FCV-10, FCD-1500 and FCV-1200 echo sounders; a CI-60G Doppler log; and A-3 GMDSS.

The owners opted for deck machinery from two manufacturers - Triplex and Rapp. The Triplex outfit includes: an 850/500/2S 30-tonne net hauler; NK-7500 six-tonne net stacker crane; KN-50 four-tonne deck crane; and KN-30 2.5-tonne fish pump crane.

Rapp provided an extensive set of winches including: three TWS-22060 58-tonne trawl winches; four SW-2500 18-tonne sweepline winches; two GW-2500 18-tonne gilson winches; two GW-2000 10-tonne outhaul winches; and a GW-2500 codend winch.

The vessel has six single berth cabins and eleven double berth cabins to accommodate a crew of up to 28.

Main Particulars

Builder: Kleven Verft, Norway
Vessel name: Vilhelm Thorsteinsson
Owner/Operator: Samherji hf, Iceland
Designer: Vik Sandvik, Norway
Total number of sister ships already completed (excluding this ship): 0
Total number of sister ships still on order: 0
Delivery date: September 2000
Length oa: 79.00m
Length bp:
Breadth, moulded: 16.00m
Depth to 1st deck: 9.5m
Depth to 2nd deck: 6.6m
Gross tonnage: 3,214 tonnes
Service speed: 18.3 knots
Bollard pull: 90 tonnes
Classification: DNV +1A1, Fishing Vessel, Stern Trawler, S, Ice C, EO
Main Engines
Model: 12V32
Manufacturer: Wärtsilä
Number: 1
Output: 5,520kW
Generators
Number: 2
Make/type: 1 x ABB AMG 560 L6, 1 x MAS 95 (844kW - emergency use)
Output (main generator): 2,500kW
Auxiliaries
Number: 3
Make/type: 1 x Mitsubishi MAS 1330 (1,091kW), 1 x MAS 580 (544kW)
Gearbox
Make: Wärtsilä
Model: SCV 95-P58
Number: 1
Propeller
Manufacturer: Wärtsilä
Number: 1
Diameter: 3,800mm
Open or nozzled: nozzled
Bow Thruster
Make: Brunvoll
Number: 1
Output: 650kW
Deck Machinery
Triplex 850/500/2S net hauler; Triplex NK-7500 net stacker crane; Triplex KN-50 deck crane; Triplex KN-30 fish pump crane; 3 x Rapp TWS-22060 trawl winches; 4 x Rapp SW-2500 sweepline winches; 2 x Rapp GW-2500 gilson winches; 2 x Rapp GW-2000 outhaul winches; 1 x GW-2500 codend winch
Capacities
Freezing hold: 1,180m³
Fish bins: 86m³
Fuel oil: 870m³
Potable water: 75m³
Ballast water: 320m³
Bridge Electronics
Radar: 1 x Furuno FAR-2835S S-band, 1 x Furuno FR-2115 X-band
Echosounders: 1 x Furuno FCV-10, 1 x Furuno FCD-1500, 1 x Furuno FCV-1200
GMDSS: A-3 GMDSS
Doppler log: CI-60G
Sonar: FSV-24
Trawl sonar: Simrad FS-925/903
Complement
Crew: 28 max.
Number of cabins: 17
Additional Equipment
Eltec ANX 95 fire alarm system; UMAS alarm, sounding and tank handling systems; Deif power management system; DSB liferafts; DSB-Kopas MOB boat; ABB main switchboard; Allweiler engine room pumps; Bombas Azcue bilge pumps; Norseligh searchlights



ARDENT & CU NA MARA: twins launched in Irish fleet renewal programme

The first two 25m vessels to be launched as part of the EU-supported Irish groundfish fleet renewal programme, *Ardent* and *Cu Na Mara* are identical vessels built for two independent Irish owners. Construction took place at Chantiers Piriou in Concarneau, France.

Measuring 24.9m x 8.0m with a 3.9m draft and weighing 233grt, each is powered by a Caterpillar 3512 EUI diesel engine delivering 615kW at 1,600 rev/min. The engines drive Renou Dardel 2,200mm diameter CP propellers with fixed nozzles through type 1/

6.054 ZF Masson ESD 701 gearboxes with triple power take-off. Auxiliary power comes from Leroy Somer LSAM461S2 alternators developing 115kVA at 1,500 rev/min.

Deck gear is extensive and comprises: three TS 2050.221B 15 tonne-pull split trawl

winches; two 2.2 ECH AC 35.48 double net drums; TCG 1980 six-tonne pull gilson winch; TAC 2P 3500 3.5 tonne pull capstan; Scantrol 2000 symmetry autotrawl system; a nine-tonne Hydro Armor hydraulic crane; and Thistle Marine type MFB14 hydraulic crane operating at 14 t/m.

A well laid out wheelhouse is packed with the latest in navigation and communication equipment from Furuno, Skipper and Scanmar.

Built to Bureau Veritas +1 3/3 (E) Deep Sea Fishing Vessel class, both trawlers operate with a crew of eight.

Main Particulars

Builder: Chantiers Pirioux, France
Vessel name: Ardent
Owner/Operator: Michael and John Orpen
Total number of sister ships already completed (excluding this ship): 1
Total number of sister ships still on order: 0
Length oa: 24.90m
Breadth, moulded: 8.00m
Depth, moulded: 3.9m
Classification: BV +1 3/3 (E) Deep Sea Fishing Vessel

Main Engine
Model: 3512 EUI
Manufacturer: Caterpillar
Number: 1
Output: 615kW at 1,600 rev/min

Alternator
Number: 1
Make/type: Leroy Somer LSAM461S2
Output: 115kVA at 1,500 rev/min

Generators
Number: 2
Make/type: 1 x Caterpillar 3304 DIT, 1 x Perkins D3-152-4

Gearbox
Make: ZF

Model: Masson ESD 701
Number: 1

Propeller
Manufacturer: Renou Dardel
Number: 1
Fixed pitch/controllable pitch: CP
Diameter: 2,200mm
Open or nozzled: Nozzled

Bow Thruster
Make: Hydro Armor type 900
Number: 1
Output: 90hp

Deck Machinery
3 x TS 2050.221B 15 tonne-pull split trawl winches; 2 x 2.2 ECH AC 35.48 double net drums; TCG 1980 six-tonne pull gilson winch; TAC 2P 3500 3.5 tonne pull capstan; Scantrol 2000 symmetry autotrawl system; 9-tonne Hydro Armor hydraulic crane; Thistle Marine type MFB14 hydraulic crane

Bridge Electronics
Radar: Furuno FR-2115 daylight viewing with ARP-26 ARPA; 1932 daylight
Sonars: Furuno CH-37 colour; FCV-292 and Skipper ES-5300 15in dual frequency colour
GPS: GP-80
Trawl sounder: Scanmar





ATLANTIC CHALLENGE: downsized ship calls for innovative thinking

The 60.00m x 14.5m mid-water pelagic trawler *Atlantic Challenge* boasts a wealth of innovative concepts and improved traditional systems. Owned by Martin Howley and partners of Killybegs, Ireland, the vessel was built in Norway at AS Eidsvik on a hull completed in December 1998 at Northern Shipyard, Gdansk, Poland. The ship is of Vik Sandvik design and fishes primarily for mackerel and blue whiting.

Much of the innovation lies in the onboard pumping and refrigeration systems devised to protect both crew and fish when pelagic bags are hauled alongside in rough seas. The vessel is believed to be the first of its size to be specifically designed for a stern pumping operation and has winches and cranes arranged aft for the purpose.

In order to adopt a stern pumping system, the yard modified a 16inch Rapp fish pump by extending the hydraulic pipe and fish hose forward. In this way the pump can be lifted into the trawl as opposed to being connected to the codend. Fish are pumped over the stern and immediately transferred to refrigerated sea water tanks.

A total of ten such tanks - each containing fast-chilling slurry ice - are positioned close to the ship's sides, an innovation which is claimed to aid vessel stability. The tanks, fitted with modern ozone technology for cleanliness, range from a central corridor. All piping is hidden underfoot and branches off to each tank.

The traditional method of manually regulating tank water flow via valves has been replaced by a PC-based system remotely operated from the wheelhouse.

Power is by means of a Wärtsilä Propac installation driving Wichmatic propeller and sterngear and Brunvoll side thrusters.

Both trawl winches, each of which pulls a maximum of 67.5 tons, are from Rapp

Hydema, as is a wide selection of other winching equipment. Deck cranes were supplied by Triplex.

The wheelhouse is dominated by Furuno electronics but also includes Kaijo Denki sonars, JRC and Simrad echo sounders and six catch sensors from Marport of Iceland.

Atlantic Challenge replaces a 70m vessel of the same name, which was also built in Norway. The owners sold their original ship when fishing quotas were cut and its large refrigerated tank capacity exceeded their needs. The shallower draft of the replacement vessel also facilitates sailing in and out of the base port.

Main Particulars

Builder: AS Eidsvik, Norway
Vessel name: Atlantic Challenge
Owner/Operator: Martin Howley and partners, Ireland
Designer: Vik Sandvik, Norway
Total number of sister ships already completed (excluding this ship): 0
Total number of sister ships still on order: 0
Length oa: 60.00m
Length bp: 51.60m
Beam: 14.5m
Depth to 1st deck: 8.85m
Depth to 2nd deck: 5.85m
Service speed: 17.5 knots
Classification: DNV +1A1 Fishing Vessel, Ice C

Main Engine
Model: 12 V32
Manufacturer: Wärtsilä
Number: 1
Output of each engine: 5,520kW

Auxiliaries
Number: 2
Make/type: Caterpillar 3412 TA
Output: 620kW at 1,800 rev/min

Generators
Number: 2
Make/type: Caterpillar SR4-596

Side Thruster(s)
Make: Brunvoll
Number: 1
Output: 558kW

Deck Machinery
Rapp Hydema: (TWS 22040 trawl winches; TWS-11031 top line winch; ND-20000B net drum; 2 x SW-7500B tail and mid line winches; SOW-500 net sounding winch; AMW-34/20-8.5 anchor winch; CF-601 capstan; GW-200B auxiliary winch; CP-2005-RH fish pump; HHR hose reel; GW-680 mooring winch); S M Triplex deck cranes

Bridge Electronics
Radar: 1 x Furuno FAR-2825 ARPA, 1 x Furuno FR-2130 S-Band, 1 x Furuno M-1832 daylight viewing
Autopilot: 2 x Robertson AP9 Mk3
Sonars: Kaijo Denki
Echo sounders: JRC and Simrad
GPS: 2 x Furuno GP-80
Radar plotter: Furuno ARP-26
Gyro: Furuno AD-100
Trawl monitoring system: Scanmar
Catch sensors: Marport
VHF radio: Sailor

Complement
Crew: 16

Additional Equipment
Swan Net mid-water trawls; Spectra bridges; Hampidjan backstraps; Thyboron trawl doors; Ulstein Tenfjord steering gear; Becker rudder; 2 x MMC Vectura vacuum pumps



ARCTIC ENDURANCE: Larger than average shrimper built for Canada

Larger than is the norm in the Arctic shrimp fleet, *Arctic Endurance* - a 69.9m x 14.6m trawler built in Norway by Soviknes Verft for Clearwater Fine Foods of Canada - entered service in the final quarter of 2000.

Expected to catch more than 6,000 tonnes of shrimp plus up to 700 tonnes of halibut a year in waters that stretch from the coast of Newfoundland to Greenland, *Arctic Endurance* has an extensive array of deck machinery, a large onboard processing factory and is capable of twin trawling. The vessel operates with two skippers and crew who normally work for two consecutive trips before change-over.

For a speed of around 16 knots, the vessel has a Wärtsilä 12V32 main engine of 6,800hp driving a 3,800mm diameter CP propeller. Two Caterpillar 3508B auxiliary diesel-driven generators, each of 910kW are installed, along with a Leroy Somer shaft alternator developing 2,500kW at 1200 rev/min. An electrically driven transverse thruster of 370kW is fitted, as is an Ulstein flap rudder with Tenfjord electro-hydraulic steering gear.

Deck equipment is extensive and mostly supplied by Rolls-Royce Ulstein Brattvaag. The hydraulic winch package includes three type DMC41.047 47-tonne trawl winches and six DSM63.017 17.5 tonne sweepline winches. Other winches include a 17.5 tonne anchor windlass, 5.5 tonne capstan, two 17.5 tonne gilson winches, a 12-tonne outhaul winch and two 12-tonne codend winches. Hydralift supplied three cranes of 2t/8m, 2t/10m and 5t/12m.

The vessel is fitted with a comprehensive shrimp grading, processing and freezing system supplied by Carnitech of Denmark. This includes three IQF tunnels (two of 18 tonnes and one of 15 tonnes capacity), five DSI 17-station horizontal plate freezers, three shrimp cookers, four shrimp graders and six Japanese packing stations. Cargo

is palletised in a specially chilled room, an unusual feature for the modern shrimp trawler.

Cargo hold capacity is 1,500m³, which is equivalent to around 550 tonnes of palletised product. Other capacities include a 300m³ hold for packaging materials, 770m³ of fuel oil and 50m³ each for ballast water and potable water. Accommodation is for 30 in a mix of single and double cabins with separate washrooms.

Furuno equipment dominates in the wheelhouse and includes radar, echo sounders and video/track plotters. Scanmar provided a hydroacoustic catch control system of SRU05-400TE type with Furuno CN24 and CN22 trawl eyes. Furuno was also sourced for an Inmarsat B and C satellite communication system, three VHF radios and a communication system that fully complies with GMDSS requirements.

Main Particulars

Builder: Soviknes Verft, Norway
Vessel name: Arctic Endurance
Owner/Operator: Clearwater Fine Foods, Canada
Designer: Skipsteknisk, Norway
Total number of sister ships already completed (excluding this ship): 0
Total number of sister ships still on order: 0
Length oa: 69.9m
Length bp: 61.8m
Beam: 14.6m
Depth, main deck: 6.0m
Depth, shelter deck: 8.8m
Service speed: 16 knots
Classification: DNV +1A1-Ice
1B - Stern Trawler - EO; hull according to Class Notation Ice A1*

Main Engines
Model: 12V31
Manufacturer: Wärtsilä
Number: 1
Output: 6,800hp

Generators
Number: 2
Make/type: Caterpillar 3508B, each of 910kW

Deck Machinery
3 x trawl winches, 2 x gilson winches, 6 x sweepline winches, 2 x codend winches, 1 x outhaul winch - all from Rolls-Royce Ulstein Brattvaag; Hydralift aft, midship and forward cranes

Capacities
Frozen products: 1500m³
Packaging materials: 300m³
Fuel oil: 770m³
Ballast water: 50m³
Potable water: 50m³

Bridge Electronics
Radar: 1 x Furuno FAR 2835S, 1 x Furuno FR2115
GMDSS: Furuno
GPS: 2 x Furuno differential, 1 x Furuno standard
Plotters: Furuno and Ocean Vision video plotters/track plotters
Echo sounders: 1 x Furuno FCV-10, 1 x FCV1500
VHF radios: Furuno
PA system: Vingtor

Complement
Crew: 30

Additional Equipment
Carnitech shrimp grading, processing and freezing system; York Refrigeration refrigeration system with Sabroe compressors; Halvorsen steam boiler; Leroy Somer shaft alternator; Ulstein flap rudder; Tenfjord steering gear; Cosmos shrimp gear with Scanmar sensors; Bridon trawl warps; Vonin halibut trawls; Scanmar hydroacoustic catch control system; Viking 6-person and 4-person liferafts



ALERT Class: PRT trio protect Alaskan waters

Three large, powerful PRTs (prevention and response tugs) - *Alert*, *Attentive* and *Aware* - are now working in Prince William Sound, Alaska, under contract to the Alyeska Pipeline Service Company. Built by Dakota Creek Industries of Washington for owners Crowley Marine Services, each vessel is on station to help prevent or speedily respond to any oil spill threat in the Sound.

Measuring 42.67m x 12.80m, the tugs are ice strengthened for year-round operation, fitted with emergency spill response equipment and have a FiFi 1 fire-fighting system. The vessels were designed by Guido Perla & Associates of Seattle and constructed to ABS Maltese Cross A1 Towing Service standard.

Powered by two Caterpillar 3612B engines, each developing 5,096hp at 900 rev/min, the vessels feature stern-mounted Aquamaster type US CP5001 azimuthing thrusters. These have 3,500mm diameter stainless steel controllable pitch propellers inside stainless steel lined nozzles. Each thruster measures about 7.8m in height and weighs around 76 tons. Bollard pull exceeds 136 tonnes and free running speed is approximately 16 knots.

In order to fulfil a varied and demanding role, the Crowley tugs have a powerful winch arrangement supplied by the Markey Machinery Company of Seattle. Central to the set-up is a type TDS-40 tow winch, driven by a GM 8V-71 diesel engine via an Allison CRT-5633 transmission. The unit can be controlled locally and remotely via the wheelhouse and has capacity for 770m of 60mm wire and a brake load of 270 tonnes. A pneumatic or manually operated brake and clutch are fitted, a 45mm gypsy installed to starboard and an auxiliary wire drum to port.

A combination winch and anchor windlass of WYWD-20-80 type is positioned at the bow with a drum capacity of 150m x 230mm circumference softline. The unit has a line monitoring system and gypsies on both sides.

Other items of deck machinery include Western Machine Works tow pins with integral hold down and a North American marine crane.

In order to accommodate specialist oil spill response teams when necessary, two single, five double and one four-berth cabin are provided. A galley, pantry, mess room, laundry, ship's office and decontamination station make up the living quarters.

The wheelhouse is well equipped with GMDSS, electronic chart display and a Simrad Robertson autopilot/gyrocompass combination. Two radar, a depth sounder and weather fax were supplied by Furuno, a GPS from North Star and three VHF radios from SEA.

The ability to respond swiftly and efficiently to any oil tanker incident is paramount and is reflected in the extensive complement of pollution control equipment. Items include two Desmi skimmer units, a pair of boom reels (each capable of holding over 600m of Kepner Sea Curtain) and two 6m long fast aluminium recovery skiffs. Up to 164m³ of recovered oil can be held in onboard tanks.

Two wheelhouse-mounted Skum monitors, fed by a pair of 1,500m³/h pumps, provide fire-fighting cover. Two foam tanks with a total volume in excess of 38m³ are installed alongside tanks for fuel (550m³) and fresh water (35m³).

Additional protective measures include a Step Warmfloor heating system for the external deck and Schuyler Rubber heavy fendering.

Main Particulars

Builder: Dakota Creek Industries, WA, USA
 Vessel name: Alert
 Owner: Crowley Marine Services, USA
 Operator: Alyeska Pipeline Service Company, USA
 Designer: Guido Perla & Associates, Seattle, USA
 Total number of sister ships already completed (excluding this ship): 2
 Total number of sister ships still on order: 0
 Length oa: 42.67m
 Breadth, moulded: 12.80m
 Design draft: 4.88m
 Service speed: 16 knots
 Bollard pull: 136 tonnes
 Classification: ABS Maltese Cross A1 Towing Service
 Main Engines
 Model: 3612B
 Manufacturer: Caterpillar
 Number: 2
 Output of each engine: 5,096hp at 900 rev/min
 Generators
 Number: 2
 Make/type: Caterpillar 3306DITA
 Output: 190kW
 Propulsion
 Stern-mounted Aquamaster US CP5001 azimuthing thrusters with 3,500mm diameter stainless steel controllable pitch propellers
 Deck Machinery
 Markey TDS-40 tow winch; Markey WYWD-20-80 winch/anchor windlass; Western Machine Works tow pins; North American crane
 Bridge Electronics
 Radar: 2 x Furuno
 Autopilot/gyrocompass: Simrad Robertson
 GMDSS: Simrad Robertson
 GPS: North Star
 Electronic Chart Display: Simrad Robertson
 VHF radios: 2 x SEA
 Additional equipment
 Arctic Step Warmfloor heating system; Kepner Sea Curtain; Skum fire monitors; Schuyler fendering



AJAX: powerful escort tug at work off Norway

The most powerful dedicated tanker escort tug in Europe and one of the most powerful vessels of its type in the world, entered service off the west coast of Norway towards the end of 2000. Named *Ajax*, the newest tug in the Ostesjo Rederi fleet, measures 41.60m x 15.90m x 6.80m, has a bollard pull of 95 tonnes and supports a sustained escort steering force of 160 tonnes at 10 knots.

Ajax was built by Astilleros Zamakona of Bilbao, Spain, in accordance with DNV regulations as an escort tug with FiFi1 and Oil Recovery certification and fully complies with the regulations of the Norwegian Maritime Directorate.

Designed by Robert Allan Ltd, naval architects of Vancouver, Canada, in extensive consultation with the owner, and in particular with Carl Johan Amundsen, of C.J. Amundsen AS, the tug incorporates a number of unique hullform developments - the result of a comprehensive series of model tests and research carried out by Robert Allan over the past several years into the performance of escort tugs. Model tests were conducted at the Institute of Marine Dynamics in St. John's, Newfoundland and at Voith Schiffstechnik's laboratories in Heidenheim, Germany to verify the predicted performance, and to refine the design of the hull and appendages.

The tug is configured to operate in the 'skeg first' direction for all escort operations, but the hullform has been refined for best possible sea-keeping capabilities in both directions. The 'aft' end includes a raised forecastle and is the direction used for running free at sea in more severe sea states, and for any rescue or long distance towing. However,

the model tests indicated that the lines at both ends offer a very dry working deck environment in up to 6m significant seas.

Power is derived from a pair of Caterpillar 3612 diesel engines, each delivering 3,800kW at 900 rev/min to a Voith cycloidal propeller, model 36GII-270 via a Voith Turbo coupling. Both main engines also drive a fire-fighting pump and a hydraulic pump via a PTO and step-up gear at the forward end. Electrical power is supplied by a pair of Caterpillar 3306B diesel generator sets, each rated 165kW at 1,500 rev/min.

The tug has a split-level accommodation and service space, orientated about the major forward working deck. The main deck area features a fully enclosed winch room, with a double-drum Karmoy winch, containing a 250m x 90mm Superline Steelite Xtra HMPE escort towing hawser and a 1,200m x 64 mm steel towline. Adjacent to the winch room is a large 'garage', containing an oil spill recovery skimmer and oil containment boom on a common bed frame. A large Effer model 80-000-3SL knuckle-boom crane serves to launch and recover this equipment as well as handle deck equipment and stores.

There is a wet lobby that accesses the crew accommodation from the main deck. The lower deck has seven well-appointed,

spacious single cabins, and a crew exercise room, all situated around a central service and storage core. The upper level includes the master's cabin, galley, galley store and roomy crew lounge and mess areas. At the boat deck level there is a large office/command centre from which any emergency response operations would be conducted. The entire accommodation lining and outfitting system, including a floating floor system was supplied and installed by Maritime Monterig AS of Bygstad, Norway. Noise levels in all accommodation areas were measured on trials at between 59 and 65 dBA under full power.

The wheelhouse is fully appointed with the latest in electronic control and navigation systems for operation of a tug with a fully automated machinery space, and includes two separate Voith steering and control consoles, one forward and one aft for maximum operational visibility and control when operating in either direction.

Main Particulars

Builder: Astilleros Zamakona, Spain
Vessel name: *Ajax*
Owner/Operator: Ostesjo Rederi AS, Norway
Designer: Robert Allan Ltd, Canada
Total number of sister ships already completed (excluding this ship): 0
Total number of sister ships still on order: 0
Length oa: 41.60m
Beam: 15.90m
Depth, moulded: 6.25m
Service speed: 15 knots
Bollard pull: 95 tonnes
Classification: DNV Escort Tug, FiFi1 & Oil Recovery Certification
Other important international/national regulations complied with: National Regulations of the Norwegian Maritime Directorate

Main Engines
Model: 3612
Manufacturer: Caterpillar
Number: 2
Output of each engine: 3,800kW at 900 rev/min

Generators
Number: 2
Make/type: Caterpillar 3306B
Output: 165kW at 1,500 rev/min

Propulsion
Voith cycloidal propeller, model 36GII-270, driving through a Voith Turbo coupling

Deck Machinery
Karmoy double-drum winch with Superline Steelite Xtra HMPE escort towing hawser;
Effer 80-000-3SL knuckle-boom crane

Capacities
Fuel: 242 tonnes
Reserve fuel: 350 tonnes
Potable water: 72 tonnes
Recovered oil: 430m³
Fire-fighting foam: 28m³
Oil dispersant: 17m³

Complement
Crew: 8
Number of cabins: 8



ENSELINI & PALMIET: on 24-hour shifts in Durban

The two Voith tractor tugs *Enselini* and *Palmiet*, delivered in the second quarter of 2000, were significant in two ways. Firstly, they added highly efficient modern tonnage to the fleet of customer and operator Portnet and also they marked a return to commercial vessel building for South African yards after 15 years of virtual inactivity.

Built by SAFReight Shipbuilding (Pty) Ltd (SAFBuild), a joint venture of SAFreight Ltd and Southern African Shipyards (Pty) Ltd, the 30.80m x 11.00m Voith Schneider tractor tugs entered service in Durban within three weeks of each other.

Utilising the *Trafalgar* class design drawn up by McTay Marine of Liverpool, UK, SAFBuild made a series of layout modifications to customise the vessels for dual 12-hour shift operations in often harsh sea conditions. Crew comfort and sound

The aft deck is dominated by a Rolls-Royce Ulstein Brattvaag winch.



insulation, for example, were major considerations. The deckhouse was specially panelled and insulation of the engine room deckhead and accommodation floor have kept the high frequency sound levels to within internationally acceptable norms.

Both the master and chief engineer have spacious separate cabins and a shared toilet/shower room. Each of the two crew cabins can sleep up to three (although two is the norm) and a large well-planned mess room has good seating arrangements. A dual shower/toilet facility is also provided for the crew and is accessed from the cross-alleyway.

Unlike other tugs in the Portnet fleet, which have Schottel propulsion systems, *Enselini* and *Palmiet* are propelled by powerful Voith 28 GII/210 units. These are powered by Ruston 6RK270 Mk1 diesels, each delivering 1,900kW at 1,000 rev/min. The engines are mounted amidships, with a Kumera Norgear gearbox on the free ends. Each has a PTO

Main Particulars

Builder: SAFBuild, South Africa
Vessel name: Enselini
Owner/Operator: Portnet, South Africa
Designer: McTay Marine, UK
Total number of sister ships already completed (excluding this ship): 1
Total number of sister ships still on order: 0
Length oa: 30.80m
Beam: 11.00m
Design draft: 5.20m
Service speed: 12 knots

Bollard pull: 58 tonnes
Classification: Lloyds +100A1 +LMC FIFI1 with Occasional Coastal Passage certification
Other important international/national regulations complied with: International Safety Requirements as set out by SA Maritime Safety Association Class 8 regulations

Main Engines
Model: 6RK270
Manufacturer: Ruston
Number: 2
Output of each engine: 1,900kW at 1,000 rev/min

Generators
Number: 2
Make/type: Cummins 6BT 5.9D (M)
Output: 91kW

Step-up Gearboxes
Make: Kumera Norgear
Number: 2

Propulsion
2 x Voith 29 GII/210 propulsors

Deck Machinery
Rolls-Royce Ulstein Brattvaag towing winch; Palfinger deck crane; Petrel Engineering anchor windlass/warping drum

Complement
Crew: 6
Number of cabins: 4

Additional Equipment
Ingersoll Dresser fire-fighting pump; Hamworthy general service pumps and Super Trident sewage treatment plant; Land and Marine air-conditioning system; Vokes fuel coalescers; Facet CPS 5B Mk 3 oil-water separator; Siemens main switchboard; Atlas Copco LT75-30KE air compressors; FW Hydrophore system; Bauer Breathing-Air bottle recharge compressor; Electrowave 98-channel engine room data logger; Counterfire fire monitors

shaft for both a 1,200m³/hr Ingersoll Dresser fire-fighting pump and a low pressure hydraulic pump for the Rolls-Royce Ulstein Brattvaag towing winch. The arrangement produces a maximum speed in excess of 12 knots and a bollard pull of around 58 tonnes.

Two 45m³/hr Hamworthy general service pumps, a Hamworthy Super Trident sewage treatment plant, Land & Marine air-conditioning system and Petrel windlass hydraulic plant complete the starboard side of the engine room.

Aft of the engine room are two watertight compartments. The port compartment houses two Atlas Copco LT75-30KE air compressors, FW hydrophore system, domestic hot water geyser and two Cummins 6BT 5.9-D (M) 91kW diesel generator sets, as well as bridge control hydraulics for the winch.

The starboard compartment holds the ropes and also gives access to the aft deck via a hatch. It also houses a shore supply transformer, Bauer breathing-air bottle recharge compressor and hydraulic pump for the Palfinger deck crane.

Forward of the Voith units, the watertight bulkhead separates a spacious workshop and control room from the engine room. Although not included in the original design, Portnet requested the inclusion of a Motor Watchkeepers' Control Booth. This position houses a locally supplied 98-channel engine room data logger, developed by Electrowave



Good visibility from a modern compact wheelhouse.

of Durban. Visibility from the booth is good and keeps the noise levels to below 77dBA in the manned compartments whilst the vessel is at normal operating power.

The wheelhouse is dominated by the Voith operating console although the steering and pitch control stands have been redesigned to allow the Master to move his seat between the consoles so as to face either forward or aft during operations. The port stand is for steering and engine control whilst the starboard stand is for pitch and towing winch control. Fire monitor operation is via a portable control box that enables the watch keeper to observe the fire from any position on the bridge deck. The new layout has greatly improved operability and reduced

crew fatigue.

Layout of the working decks is functional and clean. The boat deck (above the accommodation) houses a boat tender davit, the fire-fighting manifolds and Palfinger deck crane as well as a single 850kg stockless anchor with eleven cables of chain in an anchor pocket located on the starboard side. A Petrel Engineering anchor windlass/warping drum is positioned forward of the forward emergency escape hatch.

The aft deck is dominated by the towing winch, which has a 210 tonne brake capacity on the bottom layer and good retrieval speeds under 39 tonnes load.

For fire-fighting duties, each tug is fitted with two monitors, one for foam and the other for water. Fitted with flapper nozzles, they provide a 120m x 45m reach when both fire pumps are operating at 1,000 rev/min. Foam is stored in a 10-tonne tank.

Now providing a continuous 24-hour service in Durban and reportedly specifically requested by visiting ships, each tug is well able to handle vessels that hitherto required a two-tug team. They are not, however, restricted to harbour duties. With sufficient fuel, lube oil, fresh water and cold room storage for ten days at sea, the tugs can be detached for coastal towing or Search and Rescue duties, if necessary, at short notice.



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BUNBURY Class: first of six for operation down under

Although subjected to a troubled build programme, the first of a series of six reverse tractor tugs entered service in 2000 with Australian operator, Adsteam Marine. First of class, *Bunbury*, was loosely based on an old proven Adsteam tug design from Barnes & Fleck, an Australian firm which was asked to update and enhance its concept for the new series.

Various modifications were made. In order to improve directional stability and water flow to the propulsion units, the beam was increased and the skeg profile altered to run almost the full length of the keel, at a height of 0.5m. The changes shifted the centre of pressure forward to give improved static bollard pull and enhance indirect pull above eight knots. A thickened flat sheer strake without the traditional sponson was also incorporated and fitted with a 1,000mm Shibata fender forward and Trelleborg block fenders aft. Particular attention was paid at design stage to reducing noise and vibration levels to between 64 and 75 dB(A).

A speed of 12.5 knots and bollard pull of 60 tonnes is derived from a Daihatsu/Aquamaster power/propulsion package. Twin Daihatsu 6DKM-28 diesels, each developing 2,400hp (1,800kW) at 750 rev/min, turn stern-mounted Aquamaster US 2001 thrusters with 2,400mm diameter propellers via Twin Disc 3000-6-LD marine control drives. Auxiliary power is provided by two Caterpillar 3304DIT diesel powered gen sets, delivering a total of 170kW at 1,500 rev/min. The main machinery is cooled by four NRF type U1425 box coolers.

Featuring good all-round visibility thanks to a high rise wheelhouse with raked windows and low squat funnels, *Bunbury* has been carefully designed for a small crew. The captain and chief engineer both have

double cabins at the forward end of the deckhouse, ahead of the mess, and three twin-berth crew cabins are positioned forward below main deck. The mess area includes a separate galley, two washrooms and a laundry.

The wheelhouse is accessed via two sets of internal stairs. Compact and efficiently laid out, it has a horseshoe shaped control console forward. Individual throttles and steering controls are complemented by Aquamaster's recently developed Aquaduo single joystick control system. Electronics include a Simrad Robertson autopilot, Simrad combined GPS/echosounder, Koden radar and GMDSS pack.

Rolls-Royce Ulstein Brattvaag single split drum winches are fitted fore and aft with a certified 200 tonne proof load on the brakes and slack line speed of 0-118m/min. Both winches hold 80mm Marlow Optiline. A single stockless Bower anchor with 155m of 20.5mm galvanised studlink anchor cable is located forward.

Construction of *Bunbury* started at the Oceanfast yard but, following its closure, was completed in Fremantle, Australia by the Singapore-based company Labroy Marine. The other five in the series (*Tingari*, *Bullara*, *Beltana*, *Clontarf* and *Burra*) were transported as part-builds to Labroy's facility on Batam Island, Indonesia for completion. They are now in service at various Australian ports.

Main Particulars

Builder: Oceanfast, Australia and Labroy Marine, Australia/Singapore
Vessel name: Bunbury
Owner/Operator: Adsteam Marine, Australia

Designer: Barnes & Fleck, Australia
Total number of sister ships already completed (excluding this ship): 5
Total number of sister ships still on order: 0

Length oa: 31.88m
Beam: 11.24m
Depth, moulded: 4.72m
Service speed: 12.5 knots
Bollard pull: 60 tonnes
Classification: ABS Maltese Cross ACCU

Main Engines
Model: 6DKM-28
Manufacturer: Daihatsu
Number: 2
Output per engine: 2,400hp (1,800kW) at 750 rev/min

Generators
Number: 2
Make/type: Caterpillar 3304DIT
Output: 170kW total

Gearboxes/Drives
Make: Twin Disc
Model: 3000-6-LD MCD
Number: 2

Propulsion
Stern-mounted Aquamaster US 2001 thrusters with 2,400mm diameter propellers

Deck Machinery
Rolls-Royce Ulstein Brattvaag drum winches; Bower anchor

Bridge Electronics
Radar: Koden MD3641
Autopilot: Simrad Robertson AP9 Mk3
GMDSS: Koden
GPS: Simrad CE30 combined GPS/echosounder

Complement
Crew: 8
Number of cabins: 5



ABEILLE FECAMP: reverse tractor packs plenty of punch

The decision to upgrade its fleet of Dunkerque *Clairvoyant* class tugs for reverse tractors prompted French operator Les Abeilles to order 14 vessels of different types in 99/00 from the Alstom Leroux Naval yard in St Malo, France. The largest models are a quartet for the port of Le Havre - *Abeille Fecamp* and its sisters have large stern-mounted thrusters and achieve an impressive bollard pull of 60 tonnes. Three of the tugs are identical, the fourth - the second in the series - being larger and equipped differently for ocean-going capability.

Design aspects were influenced by the climate in which the tugs operate and also by the builder's need for a flexible layout, which could be adapted for vessels of different lengths.

Length at 36.80m and displacement (845 tons at full load) were determined by the variable winter sea state conditions encountered between Le Havre and the Antifer oil terminal, two hours north. The hull comprises a cylindrical mid-section that enables the builder to offer any length between 33m and 40m and, hence, also customise the interior. This option was put to the test for the second Les Abeilles tug, *Abeille Gascogne*, which has two additional three-berth cabins forward of the engine room to accommodate extra crew on deep sea operations.

As standard, the tugs have a full width galley/mess and two single crew cabins at main deck level and three single berth cabins on the 'Captain Deck'. Each has a separate toilet compartment.

All four vessels have identical propulsion systems. Power is derived from twin ABC 8DZC diesels with a total output of 3,680kW at 1,000 rev/min. These turn Aquamaster US 3001 thrusters with 2,600mm propellers via Twin Disc MCD 3000 slipping clutches. Free running speed is approximately 12.5 knots. A Schottel bow thruster of 200kW is also fitted, powered by hydraulic pumps that

additionally serve the towing winch system, driven off the port main engines.

The starboard engine drives the fire pump, which in the case of *Abeille Fecamp* and *Abeille Gascogne* comprises a 450m³/hr system supplying a 300m³/hr single Fishcon monitor and a self-drenching system. Vessels three and four, however, have been fitted with a Kvaerner fire-fighting system consisting of two 1,200m³/hr monitors and a drenching system fed by a pair of 1,400m³/hr main engine driven pumps.

The raised forecastle deck is dominated by a double drum 135 tonnes brake AMGC anchor/towing winch. The port drum is fitted with 100m of 52mm rope combined with 15m of 88mm nylon Sigmaflex and 20m of 44mm wire. The starboard line combines 300m of 52mm rope with a similar end arrangement. Aft of the deckhouse on three of the tugs is a single drum winch of similar characteristics, which houses an identical rope/wire arrangement to the forward starboard drum. *Abeille Gascogne* has a double drum AMGC winch aft capable of holding an extra 500m of 52mm wire.

All winches are controlled from two consoles in the wheelhouse. These also incorporate controls for the main engine and thruster and house all navigation equipment such as the Sailor VHF's, Atlas 1000 colour radar and Navitron autopilot. The tugs are equipped to GMDSS Area 1.

Main Particulars

Builder: Alstom Leroux Naval, St Malo, France

Vessel name: Abeille Fecamp

Owner/Operator: Les Abeilles, France

Total number of sister ships already completed (excluding this ship): 3

Total number of sister ships still on order: 0

Length oa: 36.80m

Beam: 10.60m

Displacement: 845 tons at full load

Design draft: 5.25m

Service speed: 12.5 knots

Bollard pull: 60 tonnes

Main Engines

Model: 8DZC

Manufacturer: ABC

Number: 2

Output per engine: 1,840kW at 1000 rev/min

Gearboxes/Drives

Make: Twin Disc

Model: MCD 3000

Number: 2

Propulsion

2 x Aquamaster US 3001 thrusters with 2,600mm propellers

Bow Thruster

Make: Schottel

Number: 1

Output: 200kW

Deck Machinery

AMGC single drum and double drum anchor/towing winches

Bridge Electronics

Radar: 1 x Atlas 1000

Autopilot: Navitron

GMDSS: Area 1

VHF radio: Sailor

Complement

Crew: 5

Number of cabins: 5

Additional Equipment

Fishcon fire monitor and self drenching system or Kvaerner fire-fighting and drenching system



FAIRPLAY 25 & 26: German duo with bulbous bows

An innovative bow design undoubtedly marks out *Fairplay 25* and *26* as unusual tug deliveries of past months. Built at the Spanish yard of Zamakona for the German owner Fairplay's business in the Dutch port of Rotterdam, the operator asserts they are superior in operation to the 1998 additions *Fairplay 21, 22, 23* and *24*.

Designed by Spanish naval architects Cintrana, each has a bulbous bow and 'deadwood' comprising a skeg, bilge keels and box keel that extends to the bow. The shape was developed in tank tests and aims to enhance course stability when working astern in reverse tractor mode.

Measuring 34.85m x 10.90m with a maximum draft of 5.20m, the tugs are powered by twin Deutz SBA 9M628 main engines, each developing 2,750hp at 1,000 rev/min. These turn stern-mounted Schottel SRP 1515 Rudderpropellers to produce a bollard pull of 66 tons and free-running speed of 13.5 knots ahead or 11 knots astern. The package also includes a Schottel type STT 110LK bowthruster with a thrust of 2.8 tonnes. Auxiliary power comes from three MAN engines, one providing 109hp and two at 240hp.

In order to meet harbour and deep sea towage requirements, along with fire-fighting and salvage duties, the tugs have a high class of accommodation, comprehensive onboard equipment and an impressive array of electronic and communications equipment, supplied by STN Atlas. The package includes a sea-chart system, GPS, two daylight radar, echosounder, gyrocompass and autopilot. The GMDSS set-up includes three VHF radios, Inmarsat C and GSM for telephone and fax communication.

A pair of double drum Rolls-Royce Rauma Brattvaag SLM 63-2T winches are fitted fore and aft, both with a brake holding load of 125 tons. The forward unit holds a total of 300m of 48mm diameter wire - divided between each drum - for harbour towage and the aft winch has 1400m of 48mm wire for ocean towage on one drum and 150m of wire

on the other. A Midguard towing hook is fitted forward and to port of the aft winch.

Although largely clear for offshore work, the aft deck houses a special pedestal for a Palfinger crane, capable of lifting 1.8 tons at an outreach of 11.4m. The forecastle deck is, however, extended aft on the port side to support a MOB boat, which is deployed by a dedicated davit.

Other principal equipment includes a FiFiI class Kvaerner Eureka fire-fighting system, powered by the main engines via Kumera-Norgear step-up boxes driving 1,350m³/hr pumps. These feed the monitors, which are positioned on a special platform spanning the funnels.

Despite meeting regulations for three-man harbour towage operations (the reason for the extensive wheelhouse fit-out) and in order to undertake ocean towage jobs, *Fairplay 25* and *26* have fully air-conditioned accommodation for up to nine crew. Three twin cabins are located at the fore end of the main deck ahead of a large mess/day room and separate galley. Large single en suite cabins are positioned on the forecastle deck for both captain and chief engineer. There is also a hospital on this level where a ninth crew member could be accommodated.

Main Particulars

Builder: Astilleros Zamakona, Bilbao, Spain
 Vessel name: Fairplay 25
 Owner/Operator: Fairplay/Richard Bouchard GmbH
 Designer: Cintrana, Spain
 Total number of sister ships already completed (excluding this ship): 1
 Total number of sister ships still on order: 0
 Length oa: 34.85m
 Beam: 10.90m
 Max draft: 5.20m
 Service speed: 13.5 knots ahead, 11 knots astern
 Bollard pull: 66 tons
 Classification: GL+ 110A5 M E Tug + MC E Aut
 Other important international/national regulations complied with: rules of the Seeverbundgenossenschaft (SeeBG) and Germanischer Lloyd
 Main Engines
 Model: SBA 9M628
 Manufacturer: Deutz
 Number: 2
 Output of each engine: 2,750hp at 1,000 rev/min
 Generators
 Number: 3
 Make/type: MAN
 Output: 1 x 109hp, 2 x 240hp
 Propulsion
 Stern-mounted Schottel SRP1515
 Rudderpropellers
 Bow Thruster
 Make: Schottel STT 110LK
 Number: 1
 Output: 2.8 tonnes
 Deck Machinery
 2 x Rolls-Royce Rauma Brattvaag SLM 63-2T winches; Midguard towing hook; Palfinger crane; MOB boat
 Bridge Electronics
 Radar: 2 x STN Atlas daylight
 Autopilot: STN Atlas
 GMDSS: STN Atlas
 VHF radios: 3 x STN Atlas
 Complement
 Crew: 9 (maximum)
 Number of cabins: 5
 Additional Equipment
 Kvaerner Eureka fire-fighting system; Kumera-Norgear step-up boxes

The bulbous bow design aids course stability.





NORTH ATLANTIC PETREL: escort skeg improves tug performance

Although at first glance identical to the *Atlantic Spruce* class design from the Canadian firm of Robert Allan Ltd, sister tugs *North Atlantic Petrel* and *North Atlantic Osprey* incorporate an unusual 'escort skeg', which has proved highly effective in enhancing towing operations.

Owned by Newfoundland Marine Energy Ltd and operated by Comtug at the Come-by-Chance oil refinery in Newfoundland, the vessels undertake escort and tanker berthing duties in high winds and sea states. Robert Allan Ltd amended the all-welded steel design to incorporate a skeg-shaped box keel from near the bow to amidships. This effectively prevents any disturbance in the water flow from affecting the stern-mounted thruster.

Reports indicate that the desired purpose of the skeg in improving course stability in following seas and whilst carrying out 'over the bow' high-speed towing operations, has been highly successful. In indirect mode, a significant improvement in line pull is evident. The skeg has not altered straight line performance and the new tugs have the same 51 tonne bollard pull and 14 knots running speed as the other eight near identical vessels built at East Isle Shipyard on Prince Edward Island, Canada, since 1995.

All ten tugs have the same power package. Twin Caterpillar 3516 diesels, each developing 2,000hp at 1,800 rev/min, turn Aquamaster US 1701 azimuthing thrusters with four-bladed fixed pitch propellers of 2,150mm diameter. The engines aboard *North Atlantic Petrel* and *North Atlantic Osprey* are, however, of the EU1 type as a result of the

availability of diesel electronic management technology.

North Atlantic Petrel has a main engine driven 2,700m³/hr water and 300 m³/hr foam FiFi1 fire-fighting system feeding two roof-mounted monitors. The foam tank holds 13,300 litres.

Accommodation comprises three single and one twin-berth cabin.

Main Particulars

Builder: Irving Shipbuilding, Inc, Canada
Vessel name: Atlantic Petrel

Owner/Operator: Newfoundland Marine Energy Ltd

Designer: Robert Allan Ltd, Vancouver, Canada
Flag: Canadian
Total number of sister ships already completed (excluding this ship): 1
Total number of sister ships still on order: 0

Length oa: 30.80m
Length bp: 29.00m
Breadth, moulded: 11.14m

Depth, moulded: 5.21m
Gross tonnage: 360 approx.
Displacement: 580 tonnes

Max speed (100% MCR engine output): 12.5 knots
Bollard pull: 50 tonnes
Range: 990nm
Classification: +100A1 Tug; LMC UMS Unlimited Service; Canadian Coast; Transport Canada; Home Trade

Main Engines
Model: 3516 EU1
Manufacturer: Caterpillar
Number: 2
Output: 2,000hp at 1,800 rev/min

Generators
Number: 2
Make/type: Caterpillar 3306 DIT
Output: 1,800 rev/min

Propulsion
2 x stern-mounted Aquamaster Z-Drives of US 1701 type. 2 x NiAl bronze fixed pitch 2,021mm propellers in nozzles

Deck Machinery
Timberland line handling winch

Bridge Electronics
Radar: Furuno FR 7062
GPS: KGP 98
Gyro: Marconi-Anschutz

Complement
Crew: 6 maximum
Number of cabins: 4

A skeg-shaped box keel is incorporated from near the bow to amidships.





STIRLING IONA: powerful AHTS at work off Shetland

Now working off the west coast of Scotland, *Stirling Iona* - a 73.89m x 16.00m anchor-handling tug supply ship - set new performance standards when it entered service in April 2000. A sister ship - intended to offer an even higher level of operational capability - was ordered by the same customer, Stirling Offshore Ltd of Scotland, in October 2000.

Built at Ferguson Shipbuilders of Port Glasgow, Scotland, *Stirling Iona* is a Vik Sandvik VS 473 design, capable of operating worldwide and well able to meet the climatic challenges in the exposed waters west of the Shetlands. The vessel has a gross tonnage of 2,528 tonnes, is registered in the Isle of Man and is classified by DNV.

Tasked for rig moving, anchor handling and platform supply duties, the ship has plenty of power with a continuous bollard pull of 173 tonnes from an installed 15,000hp; twin 12V32 Wärtsilä diesels each developing 7,500hp. These turn 3,700mm diameter Wärtsilä Propac CP propellers inside fixed nozzles. A maximum speed of 16 knots is recorded. Steering is by means of twin independent Tenfjord controlled Ulstein flap rudders with manoeuvrability further enhanced by three 816hp Brunvoll CP tunnel thrusters, one at the stern and two at the bow.

Auxiliary power is provided by two ABB 2125kVA shaft alternators, two Caterpillar 3406C diesel-driven gen sets and a Volvo TAMD 71B powered emergency/harbour set. The engines are located well forward

under the superstructure with long shafts running between the rows of cargo tanks to gearboxes aft.

The deck machinery outfit features Rolls-Royce Ulstein Brattvaag equipment comprising a windlass type B2246U for handling 1,210m of 42mm diameter chain cable and 1,845kg Pool HHP anchors; two 12-tonne pull tugger winches of AM4112 type; two 9-tonne pull capstans type CM4185; two 15-tonne pull rope reels and a triple waterfall towing/anchor winch absorbing 846kW with a main drum pulling capacity of 300 tonnes and core diameter of 2,650mm. The winch can hold 1,560m of 77mm diameter SWR, hauling at 9m/min and is capable of 400 tonnes brake load on the first layer. The towing winch is fitted with dynamic braking capacity of up to 360 tonnes and has a speed from zero to 70m/min depending on the load.

The other two drums can each hold 1,390m of 77mm diameter SWR with a brake load of 430 tonnes on the first layer and pulling capacity of 270 tonnes with dynamic braking up to 330 tonnes at speed from zero to 80m/min. The 2.4m diameter x 4.5m long stern roller was supplied by

Main Particulars

Builder: Ferguson Shipbuilders Ltd, Scotland
Vessel name: *Stirling Iona*

Owner/Operator: Stirling Ship Management Ltd, UK

Designer: Vik Sandvik, Norway

Flag: Isle of Man

Total number of sister ships already completed (excluding this ship): 0

Total number of sister ships still on order: 1

Contract date: August 1998

Delivery date: April 2000

Length oa: 73.895m

Length bp: 65.4m

Breadth, moulded: 16.0m

Depth, moulded: 7.6m

Gross tonnage: 2,528 tonnes

Displacement: 5,292 tonnes

Lightweight: 2,408 tonnes

Design deadweight: 2,884 tonnes

Design draft: 6.515m

Max speed

(100% MCR engine output): 16 knots

Service speed: 10 knots

Bollard pull: 172.5 tonnes

Range: 600nm

Fuel consumption (daily): 12 tonnes

Classification: DnV +1A1, EO Tug/Supply

Vessel SF dk(+)

Main Engines

Model: 12V32

Manufacturer: Wärtsilä

Number: 2

Output of each engine: 5,520kW at 750 rev/min

Generators

Make: Caterpillar

Number: 2

Output: 315kW at 1,800 rev/min

Gearboxes

Make: Wärtsilä

Model: SCH 85-P

Number: 2

Reduction ratio: 4.84:1

continued ...

BFS and designed for a 1000 tonne SWL to cater for the dynamic loads exerted whilst braking out the anchors. Karmoy Winch supplied four 240-tonne towpins and two 500-tonne SWL Karm forks aft to suit wire/chain sizes of 32-102mm. A 5-tonne SWL Norlift GPT100-0511 telescopic deck crane is fitted on the starboard side of the aft end of the deckhouse.

Designed to ensure maximum visibility of the winch and deck areas, the wheelhouse has a comprehensive package of electronic equipment. A main console, split either side of the helmsman's position, features a Kongsberg Simrad stand-alone SJS01 joystick system, which interfaces with the main propulsion system, thrusters, rudders and gyro. Ships Electronic Services supplied the Furuno radar, echo sounder, anemometer and DS70 doppler speed log, Robertson RGC11 gyro compass and AP9 Mk3 autopilot, GMDSS VHF, Navtex receiver, Motorola cell phone, Sailor VHF and MF/HF SSB radio telephones, Inmarsat C satcom and B satellite communications package.

A total of 965 tonnes of deck cargo can be carried whilst the Ulstein dry bulk plant, which comprises four tanks arranged along the ship's centreline, holds around 8,800ft³. The tanks can be discharged using two compressors at around 100 tonnes/hr. Other capacities include 853m³ of fuel oil, 132m³ of drill water, 427m³ of liquid mud, 534m³ of brine and 564m³ of fresh water.

As well as the normal ships bilge/ballast/fire/cooling/oil fuel and lube oil pumps, the vessel is fitted with two 100m³/hr Mission

Magnum mud pumps supplied by National Oilwell UK, a 150m³/hr drill water pump, two 100m³/hr potable water pumps, a 130m³/hr cargo fuel pump, 100m³/hr base oil pump and two 100m³/hr rig brine pumps, all from Desmi. The Dobbie McInnes cargo tank valves and above pumps are all remotely controlled in the wheelhouse and machinery control room via a Praxis alarm and monitoring system. Hamworthy supplied a Super Trident ST2A sewage treatment plant, HS 1MkII oily water separator and chockfast resins. Three Westfalia OSA-7-02-066 self-cleaning separators are supplied, one for gas oil and two for lube oil, one of which is a standby gas oil separator.

Accommodation is provided for 16, 12 in single cabins and four in double berth cabins all with private toilet facilities.

The sister ship on order (Hull No. 711) will be equipped with larger anchor handling and towing winch capacity (380 tons), increased bollard pull (about 175 tons) and chain handling capabilities. Station keeping will be assisted by three 1,000hp tunnel thrusters and a dynamic positioning system.

Glasgow based Stirling Shipping currently owns and operates a fleet of 15 offshore support vessels (12 platform supply vessels and three anchor/handling/tug/supply vessels), operating from UK and European ports in the Irish Sea, North Sea and west of Shetland. This will be the fifth vessel to be contracted over the last five years by Ferguson Shipbuilders for Stirling, bringing the total contract value to approximately £75 million.

Propellers

Material: Aluminium bronze
Manufacturer: Wärtsilä Number: 2
Fixed pitch/controllable pitch: CP
Diameter: 3.7mm
Open or nozzled: Nozzled Wageningen 19A

Bow Thrusters

Make: Brunvoll Number: 2
Output of each: 600kW - air cooled

Stern Thrusters

Make: Brunvoll Number: 2
Output of each: 600 kW - water cooled

Deck Machinery

Rolls-Royce Ulstein Brattvaag windlass, tugger winches, capstans, rope reels and triple drum waterfall towing/anchor winch; Norlift telescopic deck crane; Karmoy stern roller; Karm forks and towing pins

Bridge Electronics

Radar: Furuno
Autopilot: Robertson AP9 Mk3
GMDSS: Sailor Gyro: RGC 11

Engine monitoring/fire detection system: WECS (engine), Wormald (fire)

Complement

Crew: 16 Number of cabins: 14

Additional Equipment

Mission Magnum mud pumps; Desmi pumps; Praxis alarm and monitoring system; Pyro series 4 type E1122 oil fired boiler; Sperre Industri air compressors; Hamworthy Super Trident ST2A sewage treatment plant; Westfalia OSA7-02-066 self-cleaning separators; Wormald air-conditioning



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JAYA COMMANDER: high spec AHTS for Singapore firm

An exceptionally high level of sophistication makes the otherwise traditional AHTS ship *Jaya Commander* a significant new vessel for the year 2000. Designed by Conan Wu & Associates of Singapore and built at the Yantai Raffles Zhifu Island yard in Shandong Province, China, the vessel is now in the Singapore fleet of Jaya Holdings.

Typical AHTS features include a raised forecastle and large 400m² clear deck area aft. Special, however, is an extra upper forecastle deck which, along with main deck space, gives generous room for a much higher level of accommodation than is the norm.

Twelve single and four twin-berth cabins are apportioned to the crew whilst the captain and chief engineer benefit from suites at the forward end of the upper deckhouse. Extensive mess facilities span the full width of the ship on the main deck and six single, one twin and one four-berth cabin are provided for passengers.

Measuring 64.00m x 15.00m, *Jaya Commander* is powered by a pair of Wärtsilä 8R32 diesels, each developing 4,460hp at 750 rev/min. These are coupled to 3,500m Wichmann CP propellers turning inside Kort nozzles to ensure a bollard pull of over 105 tonnes and a maximum speed of 14 knots.

Manoeuvrability is enhanced by identical electrically driven Kawasaki type KT-72B1 bow and stern transverse thrusters. Each of these produces 515kW at 1,450 rev/min and develops 8 tonnes of continuous thrust. Electrical power is supplied by two 770kW Leroy Somer shaft generators and a pair of 300kW 3408DITA driven gen sets. A Perkins T4 236 powered emergency set of 35kW is also installed.

As would be expected, the vessel has powerful deck machinery. A 250-tonne double-drum waterfall towing/anchor handling Smatco winch, electro-hydraulically driven by a dedicated GM 12V diesel engine,

is housed in an enclosed winch room aft of the superstructure. The unit, which is remotely controlled from the wheelhouse, carries 1000m of 60mm diameter line on its upper drum and a similar length of 58mm line on the lower. The first layer has a brake load capacity of 360 tonnes.

Additional deck machinery comprises a 9-tonne anchor windlass, two 10-tonne tugger winches and two 5-tonne capstans of Plimsoll manufacture. The same company supplied a 300-tonne SWL hydraulic retractable shark jaw suitable for up to 3-inch wire or 3.5-inch chain and two sets of retractable tow pins with an SWL of 200 tonnes. A Palfinger knuckle boom crane and SOLAS approved rescue boat are also carried on deck.

Kvaerner supplied the FiFi1 BV-classified fire-fighting system. This has two 1,500m³ pumps which feed two monitors mounted on a platform between the funnels and above the wheelhouse roof. Both monitors have a 1,200m³/hr water discharge capability with a 120m throw length and one can deploy foam at 300m³/hr from a 10m³ capacity tank. A 25m³ Azcue emergency fire pump is also fitted.

Extensive cargo carrying capacity is offered on the 33m long aft deck, together with 8,000ft³ of bulk mud in four tanks, 450m³ of potable water, 10m³ of lube oil, 430m³ of drill water and 385m³ of liquid mud.

The wheelhouse extends well aft and outboard of the funnels to form enclosed bridge wings and internal access is via a central companionway. Equipment includes

a Kawasaki KICS joystick system and an extensive navigation and communications package from Furuno comprising two radar, GPS, speed log, Navtex VHF and SSB. Tokimec provided the gyrocompass and autopilot and Felcom the Inmarsat C system.

Main Particulars

Builder: Yantai Raffles, China
Vessel name: Jaya Commander
Owner/Operator: Jaya Holdings, Singapore
Designer: Conan Wu & Associates, Singapore
Total number of sister ships already completed (excluding this ship): 0
Total number of sister ships still on order: 0
Length oa: 64.00m
Breadth, moulded: 15.00m
Max speed (100% MCR engine output): 14 knots
Service speed: 12 knots
Bollard pull: 105 tonnes
Range: 7,400nm

Main Engines
Model: 8R32
Manufacturer: Wärtsilä
Number: 2
Output of each engine: 4,460hp at 750 rev/min

Generators
Number: 2
Make/type: Leroy Somer shaft generator of 770kW
Number: 2
Make/type: Caterpillar 3408DITA
Number: 1
Make/type: Perkins T4236 of 35kW (emergency set)

Propellers
Manufacturer: Wichmann
Number: 2
Fixed pitch/controllable pitch: CP
Diameter: 3,500mm
Open or nozzled: Nozzled

Bow Thruster
Make: Kawasaki KT-72B1
Number: 1
Output: 515kW at 1,450 rev/min

Stern Thruster
Make: Kawasaki KT-72B1
Number: 1
Output: 515kW at 1,450 rev/min

Deck Machinery
Smatco towing/anchor handling winch; Plimsoll anchor windlass, tugger winches and capstans; Plimsoll retractable shark jaw; Palfinger knuckle boom crane

Bridge Electronics
Radar: Furuno
Autopilot: Tokimec
GPS: Furuno
Gyro: Tokimec
VHF radio: Furuno

Complement
Crew: 20
Passengers: 12
Number of cabins: 24

Additional Equipment
Kvaerner fire-fighting system; Azcue emergency fire pump



HIGHLAND SCOUT: high spec OSV for worldwide markets

In early 2000 Bender Shipbuilding of Mobile, Alabama, USA delivered a pair of 66.20m x 16.46m offshore vessels to Gulf Offshore Marine, an affiliate of Gulfmark Offshore, with offshore operations in the North Sea, Singapore and Brazil. *Highland Scout* entered service in Brazil during mid-January and *Highland Guide* began work off Borneo some five weeks later. Conceptual design was a collaborative effort between the customer and builder with Guido Perla and Associates of Seattle assisting with the final details.

According to Gene Leech, vice-president of operations for Gulf Offshore, these are not run-of-the-mill OSVs. "We have spent a few extra dollars to give them some added dimension, with additional hardware, crew accommodation and certifications - as well as improved operational efficiencies - to increase our options in difficult international markets," he said.

On their maiden voyages, the vessels averaged 12 knots from a power plant comprising two Caterpillar 3606 diesels, each developing 2,320hp at 900 rev/min. These turn 2.5m diameter four-bladed Berg CP propellers through Reintjes gearboxes having a 4:1 reduction ratio. Steering is by Becker-style flap rudders assisted by Berg thrusters fore and aft. Both are driven by dedicated Caterpillar engines - a 1,254hp 3512 unit at the bow and a 570hp 3408C at the stern.

Between the main engines, line abreast, are three Caterpillar 3406C powered generators, each producing 250kW at 1,800 rev/min whilst a 105kW emergency set, driven by a 3304B, is located in the port aft corner of the superstructure on main deck level. Set to start automatically within seven seconds of a power failure, it will provide sufficient power to run all the navigation equipment and other crucial systems.

This deck also houses the galley, mess room, changing room, laundry and two crew cabins with a total of eight berths. Deck 1 has a day room, three four-berth cabins and

two with six berths. Captain and chief engineer have single staterooms up one more level alongside an additional two-berth cabin. Every cabin has private toilet/shower facilities. The pilothouse, which surrounds the single exhaust stack, is T-shaped in plan with control consoles fore and aft.

The vessels have an extensive cargo carrying capacity. Five bulk mud tanks along the centreline hold 210m³ and these are flanked by four liquid mud tanks each side to accommodate 639m³. Other tanks at the stern and bow, where they surround the bow thruster engine, have total capacities of 645m³ of cargo fuel oil and 1,000m³ of fresh water. Some 1,200 tonnes of deck cargo can be carried, protected by over 1.2m of freeboard.

Computer integrated pumping systems allow the chief engineer to monitor tank levels and discharge from selected tanks without leaving his control station. The vessel's entire 1,600 ton below deck hold capacity can be discharged within six hours, quadrupling the norm of 800 tons in 12 hours.

Main Particulars

Builder: Bender Shipbuilding, USA
Vessel name: Highland Scout
Owner/Operator: Gulf Offshore Marine
Designer: Customer/builder/Guido Perla & Associates
Total number of sister ships already completed (excluding this ship): 1
Total number of sister ships still on order: 0

Contract date: October 1997
Delivery date: January 2000
Length oa: 66.20m
Beam: 16.46m
Depth, moulded: 6.10m
Gross tonnage: 1830 tonnes
Deadweight tonnage: 2800 LT
Design draft: 4.91m
Service speed: 12 knots
Classification: ABS +A1 Supply Vessel, +AMS, Circle E, Loadline
Other important international/national regulations complied with: USCG Subch. L, Full Ocean SOLAS, IMO A469

Main Engines
Model: 3606TA
Manufacturer: Caterpillar
Number: 2
Output of each engine: 2,320hp at 900 rev/min

Generators
Number: 3
Make/type: Caterpillar 3406
Output: 250kW

Gearboxes
Make: Reintjes
Number: 2
Reduction ratio: 4:1

Propellers
Manufacturer: Berg
Number: 2
Fixed pitch/controllable pitch: CP
Diameter: 2.5m

Bow Thruster
Make: Berg
Number: 1
Output: 1,200hp

Stern Thruster
Make: Berg CP Tunnel
Number: 1
Output: 450hp

Capacities
Fuel oil: 790m³
Lube oil: 8m³
Potable water: 173m³
Sea Water ballast: 105m³
Fresh water ballast: 979m³
Liquid mud: 4,000 bbl
Dry mud: 8000 cu ft
Complement
Crew: 36
Number of cabins: 10



SAN RAFAEL: OSV for Mexican fields

An unusually sophisticated offshore supply vessel entered service in the Mexican oil fields in the final quarter of 2000. Ordered by Naviera Tamaulis SA of Mexico, *San Rafael*, a 62.18m x 13.41m OSV was built at the Florida yard of Eastern Shipbuilding to a design by Design Associates of New Orleans.

For operations in the Babaia de Campeche fields, off the eastern coast of Mexico, the ship is built to ABS Ocean Service classification and equipped with a dynamic positioning system and fire-fighting equipment.

Of conventional OSV layout with forward superstructure and large aft deck space measuring 42.6m x 11.9m, the vessel is powered by a pair of Caterpillar 3516B diesels, each developing 2,000hp at 1,600 rev/min. These turn 2,440mm diameter Bird Johnson four-bladed stainless steel propellers through Reintjes WAF-862 gearboxes. The rudders of the high-lift type are fitted with fishtails and have 130 degrees of travel. Manoeuvring is, however, further enhanced by a 1,000hp Thrustmaster retractable bow thruster with both azimuthing and tunnel operation. This is driven hydraulically by a dedicated Cat 3508B diesel. Five additional Caterpillar units also power three generator sets and a pair of pump sets for the four liquid mud tanks. In addition, six bulk mud tanks are provided and the vessel has extensive drill water, fuel, lube oil and potable water carrying capacity. Around 1,000 tonnes of deck cargo can also be carried.

Arguably the most important item on the bridge is the Kongsberg Simrad SDP01 dynamic positioning system with joystick, autopilot and two remote control stations. The DP is interfaced with a gyrocompass from the same manufacturer and two DGPS receivers, one by Northstar and the other from

Raytheon. This latter supplier is also responsible for two VHF's whilst Furuno supplied the bulk of the remainder of the electronics including two radar, an SSB radio, Area 3 GMDSS incorporating Inmarsat C with built-in GPS and depth meter.

The single Svenska Skum fire monitor on the pilot house roof has a capacity of 6,600 galls/min and a throw of around 140m. A 4.7m Zodiac RIB with a 40hp outboard motor is carried and deployed by a Schat-Harding davit. The vessel also has a small moonpool. Accommodation is provided for 20 in eight cabins.

Main Particulars

Builder: Eastern Shipbuilding, USA
Vessel name: San Rafael
Owner/Operator: Naviera Tamaulis SA, Mexico
Designer: Design Associates, USA
Flag: Panama
Total number of sister ships already completed (excluding this ship): 0
Total number of sister ships still on order: 0
Length oa: 62.18m
Beam: 13.41m
Classification: ABS Ocean Service
Maltese Cross A1 AMS, ABS Loadline, Full International SOLAS, International Tonnage Certificate
Main Engines
Model: 3516B
Manufacturer: Caterpillar
Number: 2

Output of each engine: 2,000hp at 1,600 rev/min

Generators

Number: 2

Make/type: Cat 3306B

Output: 170kW at 1,800 rev/min

Gearboxes

Make: Reintjes

Model: WAF-862

Number: 2

Propellers

Material: stainless steel

Manufacturer: Bird Johnson

Number: 2

Diameter: 96in

Bow Thruster

Make: Thrustmaster

Number: 1

Output: 1,000hp

Capacities

Deck cargo: 1,000 tons

Fuel: 110,000 gals

Lube oil: 1,319 gals

Drill water/ballast: 211,000 gals

Potable water: 8,636 gals

Bulk mud: 7,200ft³

Liquid mud: 3,012 barrels

Bridge Electronics

Radar: 2 x Furuno

Autopilot: Kongsberg Simrad SDP01

GMDSS: Area 3 Furuno Console,

Inmarsat C built-in GPS

DGPS: Northstar 961 colour,

Gyro: Simrad Kongsberg

VHF radios: 2 x Raytheon

SSB radio: 1 x Furuno

Fathometer: 1 x Furuno FE-680, 1 x Data

Marine

Loud hailer: SEA with 2 stations

Complement

Crew: 20

Number of cabins: 8

Additional Equipment

Skum 6,600 gpm fire monitor; Kongsberg

Simrad SDP01 dynamic positioning

system; Zodiac RB6 RIB; Schat-Harding

MOB 350/3.65/10E davit; Cat Engine

Vision System (main and bow thruster

engines)



TEAM SALALAH: AHTS built from kit cut steel

The lead vessel in a series of six, the 61.00m x 15.2m AHTS ship *Team Salalah* is unique in many ways, most particularly because it has been constructed from a steel kit.

Ordered by NICO International of Dubai in June 1998, the vessel was assembled at Adyard in Abu Dhabi to a Kvaerner Masa Marine (KMM) of Canada design utilising steel cut in the Netherlands. Delivery took place in March 2000. Work has commenced on the second vessel.

The decision to opt for a kit-build was based upon the customer's choice of yard. Adyard is essentially a repair and pressure tank fabrication facility and has little experience or equipment for a newbuild. It is, however, majority owned by NICO's parent company and was, therefore, selected on this basis.

As simplification of the build process was essential, all double curvature plating was eliminated. Shipping was also taken into consideration. The steel supplier, Centraalstaal, shipped the flat pre-formed components for each of the hull units and thus tailored dimensions to suit the size of shipping containers. The deckhouse was fabricated and outfitted in advance of the hull by Adyard itself.

KMM, which sees a strong market for kit build vessels of large size, made great efforts to ensure that modern technology was utilised to its maximum to help the inexperienced build team. Production level drawings were provided for all outfit, mechanical and electrical systems using 3D CAD modelling software to ensure best use of space and to prevent conflict between systems. Machinery and equipment were also purchased to meet KMM specifications.

A Wärtsilä NSD integrated propulsion system was selected. This comprises two 8L26 engines, each developing 2,600kW at 1,000 rev/min, driving Propac CP propellers of 2,950mm diameter in fixed Kort type 19A nozzles via Volda ACG 600H gearboxes. Brunvoll transverse tunnel thrusters, each of

430kW and developing 6.3 tonnes of thrust, are fitted bow and stern. *Team Salalah* has a bollard pull of 90 tonnes and a free-running speed of 15 knots.

Fairly standard in terms of supply vessel layout, it has a 350m² sheathed working deck area aft and 280m³ liquid mud tanks arranged amidships. Other tanks distributed along the length of the ship carry fuel oil (628m³); drill water (426m³); potable water (345m³); and bulk cement (162m³). Accommodation caters for a total of 24 persons, of which 12 are considered passengers, and includes a hospital facility.

Deck machinery, supplied by Plimsoll of Singapore, is dominated by a 120-tonne double drum tow/anchor handling winch, which is sited aft of the accommodation. Capable of pulling 159 tonnes, its upper drum holds 1,000m of 62mm cable and the lower drum is fitted with 500m of 58mm tow wire. Additional equipment includes two 10-tonne tugger winches, shark jaw and tow pins, two 5-tonne capstans, a combined anchor windlass and nylon rope storage winch and a 250-tonne SWL stern roller.

FiFi I classification requirements are met by two gearbox driven 1200m³/hr Nijhuis fire pumps, which feed two Kvaerner foam/water monitors and a Wormald self-drenching system. Auxiliary power is supplied by a pair of 580kW shaft generators backed up to two Cummins 284kW NTA855-G2M diesel gen sets and a 160kW 6CTA8.3G emergency set.

The wheelhouse is dominated by Furuno electronics including a complete GMDSS set-up with Inmarsat C, two radar, depth sounder and DGPS. The gyro and autopilot are of Anschutz type.

The remaining three ships on order will have different internal and top side configurations but will be constructed using identical hull components.

Main Particulars

Builder: Adyard, Abu Dhabi
Vessel name: Team Salalah
Owner/Operator: Nico International
Designer: Kvaerner Masa Marine, Canada
Total number of sister ships already completed (excluding this ship): 0
Total number of sister ships still on order: 5
Contract date: June 1998
Delivery date: March 2000
Length oa: 61.00m
Beam: 15.20m
Depth, moulded: 6.40m
Design deadweight: 1,725 tonnes
Design draft: 5.52m
Service speed: 15 knots
Bollard pull: 90 tonnes
Classification: ABS - A1 E AMS ACCU FiFi1

Main Engines
Model: 8L26
Manufacturer: Wärtsilä
Number: 2
Output of each engine: 2,600kW at 1,000 rev/min

Generators
Number: 2
Make/type: Cummins 284kW NTA855-G2M

Gearboxes
Make: Volda
Model: ACG 600H
Number: 2
Propellers
Manufacturer: Propac
Number: 2
Fixed pitch/controllable pitch: CP
Diameter: 2,900mm
Open or nozzled: Nozzled

Bow Thrusters
Make: Brunvoll
Number: 1
Output: 430kW - 6.3 tonnes thrust
Stern Thrusters
Make: Brunvoll
Number: 1
Output: 430kW - 6.3 tonnes thrust

Deck Machinery
Plimsoll double-drum tow/anchor winch; 2 x 10-tonnes tugger winches; tow pins; shark jaw; anchor windlass/nylon rope storage winch; 2 x 5-tonne capstans; stern roller

Bridge Electronics
Radar: 1 x Furuno FR-21355, 1 x Furuno FR-2115
Autopilot: Anschutz digital
GMDSS: Furuno
DGPS: Furuno
Gyro: Anschutz
VHF radios: Furuno FM-8500

Capacities
Deck area: 345m²
Deck cargo: 1,500 tonnes
Fuel: 992m³
Potable water: 359m³
Bulk tanks: 170m³
Liquid mud: 278m³
Refrigerator: 12m³
Freezer: 22m³
Water ballast: 24 tonnes

Complement
Crew: 12
Passengers: 12

Additional Equipment
Nijhuis fire pumps; Kvaerner foam/water monitors; Wormald self-drenching system; Jastram steering gear



GRIFFON H-181: hovercraft for Indian Coast Guard

UK based Griffon Hovercraft Ltd (GHL) completed its biggest vessel to date in mid-99 as part of a six-vessel series for the Indian Coast Guard (ICG).

The Griffon 800TD (M) hovercraft measures 21.5m x 11.30m, when its skirt is fully inflated, has a top speed of 55 knots and a payload of 8 tonnes. The craft is in use for anti-smuggling and anti-infiltration duties, particularly in areas of very shallow water around India's extensive coastline and offshore islands.



Armed with a 0.5 inch machine gun, the Griffon's capability to operate in areas difficult, if not impossible, for conventional boats, has significantly enhanced the ICG tactics. Operated by a crew of two (plus eight seamen), it can lie up and dash out to intercept over land, very shallow water, sand, rocks and mud, taking the swiftest route. Alternatively, it can patrol like conventional craft using normal berths, piers and pontoons.

The vessel is welded below the waterline, riveted and sealed above and has a 1.25m

Left: The forward raised wheelhouse seats three crew members in addition to the helmsman who has all instrumentation very close to hand.

Below: Aluminium fabrication underway at Griffon's factory in the UK.



deep skirt with trim controls estimated to have a 1,000 hour life. Although the specified speed of the lightweight alloy craft, having body measurements of 19.85m x 8.70m, is 45 knots, sea trials at launch confirmed an all-out 55 knots from a power plant consisting of two 597kW (800hp) MTU 12V 183 TB 32 water-cooled diesels, each turning a 2,600mm variable pitch propeller.

The forward raised wheelhouse has seating for three other crew members in addition to the helmsman. Down below, the fit-out is simple, to the customer's requirements, and comprises bench seating for eight with tables, a very basic windowless prison cell and a toilet compartment.

One more of the six craft on order is to be constructed in the UK with the four remaining due for assembly from knockdown kits at Garden Reach Shipbuilders and Engineers Ltd (GRSE) in Calcutta. Personnel from the Indian company attended construction courses at Griffon's factory in Southampton.

GHL, a leading player in the world of hovercraft with a 25-year history involving the design and construction of some 85 vessels, believes that local construction from kits is the way forward for significant future sales in the subcontinent. GRSE is reported as being equally confident.

Main Particulars

Builder: Griffon Hovercraft Ltd, UK
Vessel name: H-181/Griffon 8000 TD (M)
Owner/Operator: Indian Coast Guard
Designer: Griffon Hovercraft Ltd
Flag: Indian Coast Guard
Total number of sister ships already completed (excluding this ship): 2
Total number of sister ships still on order: 4
Contract date: June 1999
Delivery date: July 2000
Length oa: 19.85m
Breadth, moulded: 8.70m
Maximum all up weight: 27 tonnes
Maximum speed (100% MCR engine output): 55 knots
Service speed: 45 knots

Range: 12 hours at cruise power
Fuel consumption: 230l/hr
Classification: Lloyds +100A1 SSC HSC
LDC G3

Main Engines
Model: 12V 183 TB 32 V12
Manufacturer: MTU
Number: 2
Output of each engine: 493kW at 2,000 rev/min

Alternators
Number: 2
Make/type: 120Amp 24V per engine

Propeller(s)
Material: wood/composite
Manufacturer: Hoffmann
Number: 2
Fixed pitch/controllable pitch: CP
Diameter: 2,600mm
Open or nozzled: ducts around props

Complement
Crew: 2
Passengers: 8
Number of cabins: 0



VIGILANT Class: Australian police invest in new patrol series

Three 28.5 knot patrol boats, *Vigilant*, *Vanguard* and *Victor* have substantially increased the criminal interdiction capability of the New South Wales Police Service in Australia. Built at Image Marine in Western Australia, the craft are the lead trio in a nine-vessel order, which will comprise four further identical vessels and two larger boats at 22.00m x 5.50m.

Featuring aluminium planing monohulls from the yard's in-house design team, the vessels have a varying deadrise, with the waterline entry providing the least resistance at full operational speed whilst offering maximum comfort in varied sea conditions.

The first boats measure 16.00m x 4.88m with a draft of 1.2m, and are powered by twin Scania DI 12 42 main engines driving Teignbridge propellers through Twin Disc MG 5114 1A gearboxes. They have a maximum speed of 28.5 knots and a range of 400nm at the service speed of 18 knots. The 22m long boats have the same design speed, from a pair of MAN D2842 diesels driving via ZF BW 155 gears, and increased endurance of 600nm at 18 knots.

Internally, the fit-out is both practical and durable with two principal watertight living areas able to accommodate crews for short periods of time. Two twin berth cabins are included for the smaller vessels in the series (three cabins for the 22.00m version) and a fully equipped galley and mess with sliding cabin walls to enable the area to be opened out for certain missions.

The deck layout is equally practical and includes a towing stanchion to provide a limited towing capacity for small vessels in distress. The larger

boats carry a 4.5m semi-inflatable daughter craft, which is stowed and launched from a stern ramp recessed into the main deck and used for pursuit duties in shallow water.

Wheelhouse equipment is fairly standard to all nine vessels. All craft have 360 degree vision from the pilothouse and separate workstations for command and surveillance functions. The 16m boats have a split-level wheelhouse to further maximise all-round visibility and both classes have a flybridge with duplicate controls. These have removable windscreens and bimini tops. Onboard electronics include Raytheon R1210/6XX radar, a Raytheon L750 echo sounder, Nav 420 DGPS system, Satcom communications systems and a Saura magnetic compass.

Sergeant Graeme Jamieson, project manager for the Police Services, expressed his satisfaction with the yard's quality control and reported that the police delivery crews had remarked on the excellent seagoing ability of the vessels, especially in heavy seas. "I am confident that this service now has the technologically superior and safe platforms to substantially increase marine criminal interdiction and improve our search and rescue capability," he said.

Main Particulars

Builder: Image Marine, Australia
 Vessel name: Vigilant
 Owner/Operator: New South Wales Police Service
 Designer: Image Marine
 Total number of sister vessels already completed (excluding this vessel): 2
 Total number of sister vessels still on order: 6
 Length oa: 16.00m
 Length waterline: 14.23m
 Breadth, moulded: 4.88m
 Depth, moulded: 1.20m
 Maximum speed (100% MCR engine output): 28.5 knots
 Service speed: 18 knots
 Range: 400nm
 Classification: USL Class 2B - 200nm offshore
 Main Engines
 Model: DA 12 42
 Manufacturer: Scania
 Number: 2
 Gearboxes
 Make: Twin Disc
 Model: MG 5114 1A
 Number: 2
 Propellers
 Manufacturer: Teignbridge
 Number: 2
 Bridge Electronics
 Radar: Raytheon R1210/6XX
 DGPS: Nav 420
 Magnetic compass: Saura
 Echo sounder: Raytheon L750
 Complement
 Crew: 4
 Number of cabins: 2



CHINOOK: all-weather pilot boat serves the Bar

Oft referred to as 'the graveyard of the Pacific', the Bar - where the Columbia River collides with the Pacific Ocean - is known for its extreme weather and turbulent waters. *Chinook*, the newest pilot boat in the Columbia River Bar Pilots Association of Astoria fleet of necessity had to be swift but exceedingly durable.

Designed by Camarc Ltd of the UK and built by Kvichak Marine Industries, Inc of Seattle, USA *Chinook* measures 22.2m x 6.4m, is of aluminium construction and reportedly offers a very stable ride when transporting pilots.

The weather conditions demand that the vessel is capable of surviving a 360 degree rollover, keeping crew and equipment intact. *Chinook* was, therefore, fully tested by Kvichak on Seattle's Lake Union with the assistance of a Foss barge crane. As the crane lifted the belts, the boat was turned over 180 degrees, the straps released themselves and the vessel successfully righted. The stability test was passed due to the vessel's low centre of gravity, superstructure buoyancy and all weather double chine hull design.

Power is supplied by twin DDA 16V2000 diesels rated 1,285hp at 2,100 rev/min. These drive two Hamilton 651 waterjets

via DDA 6619 SC gearboxes with a 2.03:1 reduction ratio. During trials the boat achieved speeds of around 29 knots when fully loaded and with six crew aboard. The range has been assessed as 250nm when a speed of 25 knots is maintained.

Pilots are comfortably accommodated in four well-padded leather upholstered airline style seats from Australian manufacturer Beurteaux in a spacious cabin. A day galley with microwave, refrigerator, freezer, sink and storage is incorporated. Good visibility is ensured by a total of 28 windows, supplied by Freeman and specified as 0.5inch tempered glass with bolted aluminium frames. A cabin aft has two bunks and lockers and a separate shower/toilet cubicle alongside.

The wheelhouse area has raked windows, a full width console and mirrored control stations to port and starboard. Onboard

electronics include two Furuno 2115 console X-band 10 target ARPAs; Furuno GP80D GPS units; a Robertson AP9 autopilot and RGC10 RSR8805 repeater gyro; three SEA 156H VHF's; and two Midland 70-1340B VHF's.

Various measures have been taken to afford enhanced protection for both the vessel and its crew. MOB recovery equipment includes a customised articulated platform at the stern and an articulated A-frame hoist amidships. A Hadrian safety rail system is installed and shock absorbing foam fendering with polyethylene contact surface is fitted around the vessel. Heavy-duty welded aluminium handrails have also been specified, the forward rails being heated, and one-inch thick rubber matting is in place at the boarding area.

Electrical needs are met by six 8D and four 4D sealed batteries for 24VDC supply and two 32kW generators. Aquasignal Series 70 navigation lights and 1000W floodlights and two ACR remote spotlights are fitted.

Main Particulars

Builder: Kvichak Marine Industries, USA
Vessel name: *Chinook*

Owner/Operator: Columbia River Bar Pilots Association
Designer: Camarc Ltd, UK
Flag: USA

Total number of sister ships already completed (excluding this ship): 0

Total number of sister ships still on order: 0

Length oa: 22.2m

Breadth, moulded: 6.4m

Displacement: 48.5 tonnes

Design draft: 1.1m

Max speed (100% MCR engine output): 29 knots

Range: 250nm @ 25 knots

Main Engines

Model: 16V2000

Manufacturer: Detroit Diesel Allison

Number: 2

Output of each engine:

1,285hp @ 2,100 rev/min

Gearboxes

Make: Detroit Diesel Allison

Model: 66195C

Number: 2

Reduction ratio: 2.03:1

Waterjets

Make: Hamilton 651

Number: 2

Deck Machinery

A-frame hoist

Bridge Electronics

Radar: 2 x Furuno 2115

Autopilot: Robertson AP9

GPS: 2 x Furuno GP8-D

Gyro: Robertson RGC10

VHF radios: 3 x SEA 156H,

separate channels; 2 x

Midland 70-1340B

Complement

Crew: 2

Passengers: 4 pilots

Number of cabins: 1

Additional Equipment

Hadrian safety rail system;

Aquasignal navigation lights

and floodlights; ACR spotlights



SAN FRANCISCO: standing by the Golden Gate

Traditionally renowned for its impressive output of tugs and fishing vessels, MARCO Shipyard in Seattle, USA, has delivered the 31.7m *San Francisco* to the San Francisco Bar Pilots. The first of two new pilot station vessels outfitted with the latest technology, the pair replace existing 25.9m boats. Designed by the yard, the boats transfer pilots safely to and from ships entering or leaving San Francisco Bay, and provide quarters for the pilots between assignments. One remains on duty outside the Golden Gate Bridge at all times and under all conditions.

This type of operation demands a different style of craft to the current trend of smaller, lighter and faster pilot boats and is reflected in the accommodation. Aft the raised wheelhouse, the main deck features a spacious day room/lounge area with large windows, an entertainment and information centre, and a full galley. Armchair seating is provided for nine persons. On the lower deck, accessed directly from the lounge, is an eight-person berthing compartment for pilots forward, complete with two heads and a shower. Crew quarters for five are located aft of the amidships engine room. Sound levels have been kept to a minimum throughout all working and living spaces on the vessel. Comfort for all is enhanced by a

large roll stabilisation tank located below the pilothouse deck and large bilge keels on the hull.

Measuring 37.1m x 9.0m with a draft of 2.7m, the all-steel vessel is powered by a pair of Caterpillar 3508B diesels, each delivering 1,100hp to 1,630mm diameter Sound Propeller Ni-Al-Bronze propellers through Twin Disc MG-5301 gearboxes with a reduction ratio of 4.06:1. The new vessel exceeded its 14-knot design speed during extensive sea trials in Puget Sound and delivery trials on San Francisco Bay. The boat was accepted from its builders by Captain Peter McIsaac, president of the San Francisco Bar Pilots. "We were delighted with the performance of this boat on her trials," he

said, "and we look forward to placing her in service as soon as possible. She met our expectations, and I know the pilots will enjoy the new level of comfort and safety these boats will bring to the Bay," he concluded.

Exceptionally good all-round visibility is offered from the wheelhouse, which has direct internal access from the pilots' lounge and doors to bridge wings which have extending platforms to increase the boarding options available. Harris Electric of Seattle provided distribution panels, a master alarm panel, wiring and electronics. These include two Furuno FR21 1 5BB X-Band radar, two Furuno GP80D GPS receivers, a Furuno FCV-29 colour depth sounder, a Transas chart plotting system with Ocean PC hardware, four SEA 156 VHF radios, a Sperry MK37 VT gyrocompass, a Litton LMP HSC/GM autopilot, a Harris HelmAlert watch alarm, and a Ray 430 loudhailer.

At the stern, the vessel carries a 4.8m Ocean Tech RIB powered by a 100hp diesel engine and waterjet, which can be used as an emergency rescue boat and for pilot transfers with low-freeboard vessels. The RIB is carried on a hydraulically powered cradle that lowers the boat for release while underway, and raises it to main deck level for stowage.

Main Particulars

Builder: MARCO Shipyard Seattle, USA
Vessel name: San Francisco
Owner/Operator: San Francisco Bar Pilots
Total number of sister ships already completed (excluding this ship): 0
Total number of sister ships still on order: 1
Length oa: 31.7m
Breadth, moulded: 9.0m
Design draft: 2.7m
Service speed: 14 knots

Main Engines
Model: 3508B
Manufacturer: Caterpillar
Number: 2
Output of each engine: 1,100hp

Gearboxes
Make: Twin Disc
Model: MG-5301
Number: 2
Reduction ratio: 4.06:1

Propellers
Material: Ni-Al-Bronze
Manufacturer: Sound Propeller
Number: 2
Diameter: 1,630mm

Bridge Electronics
Radar: 2 x Furuno FR21 1 5BB X-Band
Autopilot: Litton LMP HSC/GM
GPS: 2 x Furuno GP80D
Depth sounder: Furuno FCV-29
Chart plotter: Transas with Ocean PC hardware
Gyro: Sperry MK37 VT
Watch alarm: Harris HelmAlert
Loudhailer: Ray 430

Complement
Crew: 5
Passengers: 9

Additional Equipment
Harris Electric distribution panels; Ocean Tech RIB





SPEEDY: Australian-built monohull for Europe

Having celebrated its 15th anniversary in 1999 with the completion of its 60th vessel, Western Australia-based WaveMaster International welcomed in the new millennium with the completion, on schedule, of the country's first large aluminium monohull ferry destined for service in Europe and the biggest ever built by the yard. Named *Speedy*, the 50m vessel carries up to 324 passengers between Bremen, Germany, and the Friesian Islands in the livery of Reederie Warrings - making the 80nm crossing at a speed of 30 knots.

The ferry measures 49.95m x 9.00m with a draft of 2.00m and air draft of 6m (restricted by bridges on its route). Power is derived from two Caterpillar 3516B main engines, each developing 2,694hp at 1,855 rev/min, turning 1,220mm diameter five-bladed highly skewed Teignbridge propellers through ZF BW 755 marine gears. Speed at 100 per cent MCR is 30 knots at 32 tonnes deadweight in Beaufort 2 conditions. With a fuel consumption of 198g/kWh, this gives a range of 200nm. Electrical supply is provided by two Caterpillar 3304B generator sets, each delivering 85kW at 1,500 rev/min.

All passenger accommodation is 'tourist class' on the main deck equipped with Beurteaux 'Ocean Tourist' low back non-reclining seats having aluminium arm rests, lifejacket pockets and magazine holders. These are laid out amidships in a 3-6-3 formation tapering down to 2-4-2 forward. Kiosks are provided for dispensing drinks and snacks. A zoned system of air conditioning and heating is used to optimise the temperature across the entire seating area. The toilets are located aft where any noise and vibration from the 'business end' of the vessels will have the least effect. However, in areas where seating is situated over machinery spaces, a floating deck has been incorporated.

To reduce height, the wheelhouse, which sits over a service void, projects for only half of its height above the main saloon roof and the mast can be lowered by remote control. For close-quarter manoeuvring, open

bridge wings are available on both sides.

To counter the problems of passenger discomfort resulting from the notoriously lumpy seas in the operational area, a Maritime Dynamics ride control has been installed. This comprises four flaps aft responding automatically to pitch and roll motion sensors located throughout the vessel.

Construction of the vessel began in late November '99 and this was followed, in early January 2000 by the successful completion of exhaustive tank testing at the Schiffbautechnische Versuchsanstalt in Wien GmbH (Vienna Model Basin). Self-propelled seakeeping tests were carried out, witnessed by teams from both WaveMaster and Reederie Warrings, in which North Sea conditions were replicated. These were conducted in significant wave heights of two metres at speeds of up to 27 knots in head and beam seas and at 30 degrees to the bow. The model was subjected to both regular and irregular wave patterns. At the conclusion of these tests, Richard Anzbock, deputy director of the basin, said that he was impressed with the seakeeping characteristics and predicted that it would be ideal for North Sea operations.

The launch was completed at the very end of April, following an hour and a half journey on a heavy lift vehicle from the yard's production hall to the launch site. Immediately after the launch, *Speedy* began a demanding four-week schedule of sea trials before final acceptance and delivery to Germany.

Main Particulars

Builder: WaveMaster International, Australia
 Vessel name: *Speedy*
 Owner/Operator: Reederie Warrings
 Designer: WaveMaster International
 Total number of sister ships already completed (excluding this ship): 0
 Total number of sister ships still on order: 0
 Delivery date: April 2000
 Length oa: 49.95m
 Length waterline: 42.50m
 Beam: 9.00m
 Hull draft: 1.50m
 Maximum draft: 2.00m
 Maximum air draft: 6.00m
 Max speed (100% MCR engine output): 30 knots
 Range: 200nm
 Classification: HSC Code and Germanischer Lloyd +100A5 HSC Passenger OC3
 Main Engines
 Model: 3516B
 Manufacturer: Caterpillar
 Number: 2
 Output of each engine: 2,694hp at 1,855 rev/min
 Generators
 Number: 2
 Make/type: Caterpillar 3304 BT
 Output: 106kVA each
 Gearboxes
 Make: ZF
 Model: BW 755
 Number: 2
 Propellers
 Manufacturer: Teignbridge
 Number: 2
 Diameter: 1,220mm
 Complement
 Passengers: 324
 Number of cabins: 0
 Additional Equipment
 Beurteaux Ocean Tourist passenger seats; Maritime Dynamics ride control system; Australian Air Conditioning Services system



FLYING CLOUD: high speed cat on Nantucket routes

Derektor Shipyards of New York, USA, utilised a UK design from Nigel Gee Associates to build *Flying Cloud*, a 41m long aluminium catamaran for Martha's Vineyard and Nantucket Steamship Authority, Woods Hole.

Designed for year-round passenger services to the islands of Nantucket and Martha's Vineyard, the vessel has been constructed to meet USCG Subchapter K full compliance and carries 300 passengers at a service speed of 36 knots. The wake-wash generated by the hull design is claimed to be extremely low, even at high speeds and the journey time on the Hyannis Port to Nantucket route is less than half that of competitive traditional ferries.

Power is derived from two Paxman 12 VP 185 diesel engines driving Kamewa 71 SII waterjets via Reintjes gearboxes. The jet intake tunnels are made of glass fibre to minimise corrosion problems and ensure long, trouble-free working life. Passenger comfort is enhanced by a Maritime Dynamics ride control system, which comprises a T-foil and interceptor on each hull for a smooth ride in sea conditions up to sea-state 4.

The main deck seats 171 passengers in airline quality seats and features a full service bar and four toilets aft. The upper deck has 129 seats with a central lounge area and small

bar and exterior seating is provided aft to give passengers the option of open-air travel. Both decks are fitted with a four-season HVAC system to maintain constant temperatures and eliminate window fogging.

As *Flying Cloud* is a bow-loading vessel, it has a large cargo storage area in the bow capable of holding up to 9,000lb of luggage. The area can be fitted out with bicycle racks and wheeled carts to facilitate handling of passenger luggage.

A raised pilothouse has all-round visibility and is equipped with the latest electronic and navigation aids. The package includes two Raytheon ARPA units; a Maritime Dynamics autopilot; Northstar 941XD DGPS; Sperry compass; Transas chart plotter; Vistar night vision system; and two ICOM M-127 VHF's.

In recent years Derektor has established itself as a leading builder of high-speed craft in North America. It enjoys a reputation for very high standards of aluminium fabrication and machinery installation and a weight-conscious approach to vessel construction.

Main Particulars

Builder: Derektor Shipyards,
Mamaroneck, NY
Vessel name: Flying Cloud
Owner/Operator: Woods Hole, Martha's
Vineyard & Nantucket Steamship
Authority
Designer: Nigel Gee Associates Ltd, UK
Flag: USA
Total number of sister ships already
completed (excluding this ship): 0
Total number of sister ships still on
order: 0
Contract date: May 1999
Delivery date: May 2000
Length oa: 41.00m
Length bp: 35.60m
Breadth, moulded: 10.50m
Design deadweight: 38.5 tonnes
Design draft: 1.90m
Service speed: 36 knots @ full load
Classification: USCG Subchapter K

Main Engines
Model: 12V P185
Manufacturer: Paxman
Number: 2
Output per engine: 2500kW @ 1,922 rev/
min

Generators
Number: 2
Make/type: Northern Lights M75C

Gearboxes
Make: Reintjes
Model: WVS930
Number: 2

Waterjets
Make: Kamewa 71 SII
Number: 2

Bridge Electronics
Radar: 2 x Raytheon ARPA C2525/7X
with 40 rev/min antenna
Autopilot: Maritime Dynamics
GPS: Northstar 941XD DGPS
Compass: Sperry MK37
Chart plotter: Transas
VHF radios: 2 x ICOM M-127

Complement
Crew: 5
Passengers: 300

Additional Equipment
Maritime Dynamics T-foil and interceptor
ride control system

Left: The latest electronics in the raised pilothouse. Below: Passenger seating is comfortable but practical.





THE LYNX: biggest aluminium ship to be built in Australia

Launched as *Incat Tasmania*, *The Lynx* is already well known for its starring role at the Sydney Olympics. The first vessel in the Evolution 10B class from Incat Australia, the ferry is currently on long term charter to New Zealand operator Tranz Rail.

The Evolution 10B class extends Incat's 10 year history in the production of high-speed wave-piercing ferries and delivers proven technology in a vessel capable of carrying a mix of passengers and freight at speeds over 40 knots. *The Lynx*, measuring 98m x 26m is the largest aluminium ship built to date in Australia.

Designed to accommodate up to 900 persons, the passenger area is on one deck and comprises three lounge areas. Each features its own style of seating, colour scheme and facilities. Passengers travelling without cars enter the ship through large side entry doors into the central Midships Lounge, the focal point for many onboard activities as it is surrounded by the shop, kiosk and outside seating areas.

Travellers can also sit in the Forward Lounge, which boasts 300 degree panoramic views or, if travelling with cars, in the Aft Lounge. Disability access is second to none due to a new ramp system, which ensures that the entire passenger area is wheelchair accessible.

The vessel's vehicle deck offers a total of 380 truck lane metres, 3.1m wide x 4.3m clear height and 360 car lane metres at 2.3m width x 2.1m height. Suitable for heavy road

transport vehicles, the main vehicle deck also benefits from unrestricted height of 6m and an unobstructed width of 18m at the stern.

Nine hoistable mezzanine vehicle decks give the operator the flexibility to carry 260 cars and no heavy vehicles or 12 road freight trailers with 180 cars, or 24 road freight trailers with 85 cars while offering the headroom demanded by oversized freight vehicles when raised. The vehicle decks, with clear lane markings painted on the deck, ensure fast vehicle loading and unloading with the flexibility to alter the configuration on a voyage to voyage basis.

The raised wheelhouse is small for a vessel of its size. Large windows provide 360-degree visibility for the officers over the new aerodynamic superstructure, while an aft-facing docking console and CCTV monitors negate the need for bridge wings with their associated windage. Consistent with all recent Incat vessels, the control station is fitted with the latest in electronic, navigation and communication equipment to comply with High Speed Craft Code Sea Area A2.

Unlike Incat's first three Evolution 10 Class vessels, which had Caterpillar 3618 diesels installed, *The Lynx* is powered by four Ruston 20RK270 medium speed diesels

developing in excess of 28,000kW. The engines drive transom-mounted steerable Lips 120E waterjets via Reintjes VLJ6831 gearboxes. All four waterjets are configured for steering and reversing while an independent hydraulic system in each hull covers the steering and reverse functions. A new feature is the introduction of the new Lips six-bladed, high-speed, high pressure waterjet incorporating a completely new control arrangement. Developed jointly by Incat and Lips, the new hydraulic cylinder layout is claimed to eliminate much of the maintenance and operational problems experienced by Incat on previous waterjet models. An additional benefit of the new jet is a big reduction in weight compared with the earlier 150D units, providing a saving of around 30 tonnes. The new E series waterjets will also form the basis of any new Incat gas turbine powered vessels in the future.

Another major improvement over earlier Incat vessels can be seen in the main engine dry exhaust system. First seen on Incat 053 *Bentayga Express*, the new system propels engine exhaust away from the vessel via twin funnels on top of the aft superstructure instead of into the tunnel between the vessel's hulls. Developed in co-operation with Incat's main engine suppliers Caterpillar and Ruston, the objective of the new exhaust system was to meet new international noise and emission standards. Early tests demonstrated that it meets noise level requirements set out by the Danish Maritime Authority for the Incat 91m vessels *Mads Mols* and *Max Mols*, currently trading between Aarhus and Odden. The system has also been designed to take catalytic converters, improving already efficient emission controls.

To further improve sea keeping qualities, *The Lynx* is fitted with a Maritime Dynamics/Incat Ride Control System consisting of transom mounted trim tabs and a new retractable bow T-foil. These, combined with a new hull form featuring longer outer bows, fuller midship sections, stern skeglets and improved centrebow clearances, means motion sickness incidence has reduced by up to 40 per cent in higher sea states by reducing pitch, roll and heave, the major contributors to passenger discomfort.

The vessel is fitted with six evacuation stations. Two of these, on each side of the vessel, contain an IMO approved MES (Marine Evacuation System) supplied by Liferaft Systems Australia, with an additional liferaft access station on each aft mooring deck. An MES consists of an inflatable slide, which connects with multiple 100-person liferafts. The evacuation arrangement has proved capable of evacuating the full vessel's complement in times much faster than the IMO requirements. In addition, the vessel is designed and built with high levels of reserve buoyancy, fire detection/protection and safety systems redundancy.

Hobart-based company Colbeck & Gunton supplied a lightweight structural fire protection system including fire doors and dampers. The 'Rapid Access' (deckhead) and 'Lightweight' (bulkhead) system meets all the demands for lighter weight and faster installation/removal for this type of craft. An addressable fire detection system, CCTV cameras, zoned fire sprinkler systems and

hydrants protect engine rooms, vehicle decks and passenger areas. *The Lynx* is also fitted with portable fire extinguishers, fire-protection suits and equipment, water fog applicators, breathing apparatus, international connections and fire control plans to meet IMO requirements.

Main Particulars

Builder: Incat Tasmania
Vessel name: *The Lynx*
Owner/Operator: Tranz Rail
Designer: Design@Incat
Flag: New Zealand
Total number of sister ships already completed (excluding this ship): 3
Total number of sister ships still on order: 0
Length oa: 97.22m
Length bp: 92.00m
Breadth, moulded: 26.60m
Gross tonnage: 6,581 tonnes
Displacement: 1,800 tonnes
Lightweight: 1,020 tonnes
Design deadweight: 750 tonnes
Design draft: 3.42m
Max speed (100% MCR engine output): 46.5 knots
Service speed: 38 knots
Range: 38nm
Fuel consumption (daily): 5.8 tonnes/hr at 100% MCR
Classification: DNV +1A1 HSLC R1 Car Ferry "B"
Other important international/national regulations complied with: IMO HSC Code and applicable IMO regulations
Main Engines
Model: 20RK20

Manufacturer: Ruston
Number: 4
Output of each engine: 7,080kW at 1,030 rev/min

Alternators
Number: 4
Make/type: Caterpillar
Output: 240kW at 1,500 rev/min

Gearboxes
Make: Reintjes
Number: 4

Waterjets
Make: Lips 120E
Number: 4

Deck Machinery
6 x hydraulic mooring capstans; 1 x galvanised steel anchor

Bridge Electronics
Radar: Bridgemaster X Band
Autopilot: Lips
GMDSS: Sea Area 1 & 2
GPS: Leica differential
Chart plotter: Transis
Gyro: Anschutz

Complement
Passengers: 900
Number of cabins: 0

Vehicles
No. of vehicle decks: 1 x main deck plus mezzanine decks
Total lane length: 380 truck lane metres plus 80 cars
No. of cars: 267

Additional Equipment
Maritime Dynamics active ride control system; Liferaft Systems Australia MES; Colbeck & Gunton structural fire protection system; Sanyo reverse cycle air-conditioning

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HIGHSPEED 2: fast ferry for Greek islands

A special high-speed steering system and purpose designed mooring and anchor windlasses added innovative touches to *Highspeed 2*, the first of three 'Auto Express' ferries ordered by Greek operator, Minoan Flying Dolphins Maritime, from Austal Ships in Australia.

The 72.00m x 17.50m catamaran, which entered service in and around the Aegean Islands in the summer of 2000, has capacity for 620 passengers and 70 cars. Notably, it is the first Austal vessel to be fitted with the Seastate T Max high-speed steering system, fitted as an alternative to traditional waterjet bucket steering, which the company believes results in considerable additional drag in rough seas.

T Max solves the problem by taking

Controls are literally to hand in the wheelhouse.



Main Particulars

Builder: Austal Ships, Australia
 Vessel name: Highspeed 2
 Owner/Operator: Minoan Flying Dolphins, Greece
 Designer: Austal Ships
 Flag: Greek
 Total number of sister ships already completed (excluding this ship): 1
 Total number of sister ships still on order: 0
 Delivery date: May 2000
 Length oa: 72.0m
 Length bp: 63.5m
 Breadth, moulded: 17.5m
 Depth, moulded: 5.9m
 Gross tonnage: 2,869 tons
 Displacement: 770.30 tons
 Lightweight: 580.30 tons
 Design deadweight: 190 tons (max)
 Service speed: 40 knots
 Range: 470nm
 Fuel consumption (daily): 3 tonnes/hr
 Classification: GL Maltese Cross 100 AS
 HSC-B OC3 High Speed Passenger Craft/
 Ro-Ro Type. Machinery Maltese Cross
 MC, HSC B
 Other important international/national regulations complied with: Hellenic Ministry of Merchant Marine (EEP) Requirements

Main Engines

Model: 16 V 595 TE 70L
 Manufacturer: MTU
 Number: 4
 Output of each engine: 3,866kW at 1,750 rev/min

Alternators

Number: 4
 Make/type: MTU 6R183 TE 52
 Output: 228kW at 1500 rev/min

Gearboxes

Make: Reintjes
 Model: VLJ 2230
 Number: 4
 Reduction ratio: 2.548:1

Waterjets

Make: Kamewa 90 S11
 Number: 4

Deck Machinery

John Honey hydraulics; JHH 2363 MW 7T mooring winch; JHH 2363 AW 6.7T anchor winch; Schat Harding davit

Bridge Electronics

Radar: Kelvin Hughes Nucleus 5000 II
 Autopilot: C-Plath Navipilot
 GMDSS: SP Radio
 GPS: Leica MX 412B DGPS
 Chart plotter: Kelvin Hughes Nucleus 5000 II ECDIS
 Gyro: C-Plath

Complement

Crew: 24
 Passengers: 620
 Number of cabins: 0

Vehicles

No. of vehicle decks: 1
 No. of cars: 70

Additional Equipment

Seastate ride control; Seastate T Max high-speed steering system

control of the vessel's waterjets at speeds in excess of 25 knots, allowing all waterjet thrust to be vectored for maximum propulsive thrust and eliminating bucket drag. Seastate estimates suggest that average service speed improvements of over two knots can be achieved in rough sea operations. A quadruple engine/waterjet propulsion package is installed, comprising MTU 16V595 diesels, each developing 3,866kW at 1,750 rev/min and Kamewa type 90 SII jets to give a service speed of 40 knots.

To enable passengers to board *Highspeed 2* in a safer manner, passenger-loading gangways are hinged off the waterjet guard at the aft end of each hull. Of a fixed length, these are raised and lowered hydraulically, and designed to operate simultaneously with, but independently of, the vehicle ramp.

The craft's mode of 'stern-to' berthing also presented the yard with a challenge since the

Greek islands are notorious for gusty and variable winds during the summer when there is maximum pressure to maintain schedule. This has been solved by utilising a combination of specially designed mooring and anchor windlasses and high specification anchors to ensure rapid deployment and recovery.

With 1250m² devoted to passengers and support services, the vessel is outfitted to a high standard. Amtico flooring incorporating an 'oak timber' look is used extensively throughout heavy traffic areas, creating a stylish yet durable finish. On the main passenger deck, a wide range of facilities is provided and includes two kiosk/restaurant areas, a children's playroom and a games room with several arcade machines. The shopping area and information counter is located amidships and ample baggage racks are provided.

Reclining seats, supplied by Beurteaux Australia, are featured throughout the main deck in a combination of airline and table arrangements. The business class seating area for 128 passengers is located forward on the main deck with panoramic windows across the full width of the superstructure.

The upper VIP lounge has deluxe leather seating for 80

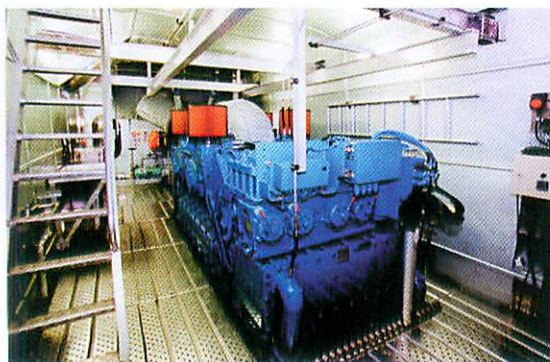
Power is derived from four MTU diesel engines.

with sound systems in the armrest. An exclusive appointment to this lounge is the inclusion of 24 individual game systems mounted in the seats - similar to those now widely used in the airline industry. It is also equipped with a bar/kiosk and generous baggage racks. A covered outdoor seating area for 28 passengers is also available aft on the main deck.

The vessel will operate with 24 crew and has extensive crew facilities. Captain and engineer are accommodated in two cabins, located forward on the bridge deck, each fully air-conditioned and equipped with separate en-suite facilities. Two twin berth cabins with shared toilet and shower facilities are adjacent to the crew mess. Electronics in the wheelhouse include Kelvin Hughes radar, navigational equipment and electronic chart navigation, C Plath gyro and magnetic compasses, Skipper echo sounder, Sailor and ICOM communications systems and Vistar night vision system.

Two marine evacuation systems from Lifteraft Systems Australia, are located forward, to port and starboard. Harbour deployment trials successfully demonstrated the evacuation of 184 passengers in under 11 minutes, well inside the required time of nearly 18 minutes.

Highspeed 2, which is fitted with a Seastate ride control system, comprising T-foils in the bows and interceptors on the transom of each hull, is to be followed by identical sister ship, *Highspeed 3*, and the larger capacity *Auto Express 92*, *Highspeed 4*.



From concept to reality . . .



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Motion Control Solutions

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AREMITI 4: Flying Cat for Tahiti

One of the earliest tasks in 2000 for Damen Shipyards Singapore Pte Ltd - the company newly formed by the Dutch Damen group after its take-over of Kvaerner Fjellstrand in Singapore - was to find a buyer for a 49.00m x 12.50m Flying Cat 50 fast ferry, completed but unsold.

Originally ordered by a Maltese customer, the vessel - to become *Aremiti 4* - underwent substantial modifications in order to meet the needs of a new customer. Damen set about major alterations to the brand new vessel by increasing its passenger carrying capacity in the main saloons from 408 to 455. This involved removing the crew accommodation, adding seating for 50 on an upper sundeck, providing a new portside access and making structural alterations to the hull bottom to satisfy French Administration damage stability requirements.

The end result meets IMO HSC Code Category B requirements and is classed with Germanischer Lloyd.

Capable of speeds of 37.5 knots, *Aremiti 4* is powered by four MTU 16V 4000 M70 diesels, each driving a Kamewa 63 SII waterjet via a ZF BW 755-ID gearbox. A Kvaerner motion dampening system is fitted, comprising forward T-foils and stern trim flaps.

Interior accommodation, as specified by owner Aremiti Pacific Cruises, provides for 150 passengers in a main deck forward saloon and 295 passengers in an upper deck saloon. The former also houses a bar/kiosk and five

toilets and the latter a bar aft and five toilets forward. Seating is of the Eknes Transit Super Nova 1410 type for the saloons and Bermuda 110 type from Beurteaux Australia on the sundeck.

Vehicles are carried on the main deck aft of the passenger saloon. Accommodation for up to 20 cars, each allotted 4.30m x 2.25m of space with 2.30m clearance, is accessed by means of two ramp doors. These have been designed to berth against shoreside facilities of between 2.25m and 3.40m above waterline and are positioned at the transom and starboard sides at the forward end of the deck.

Main Particulars

Builder: Damen Shipyards
Singapore Pte Ltd
Vessel name: Aremiti 4
Owner/Operator: Aremiti Pacific Cruises
Designer: Kvaerner Fjellstrand
Total number of sister ships already completed (excluding this ship): 0
Total number of sister ships still on order: 0
Length oa: 49.00m
Beam: 12.50m
Design draft: 1.95m
Service speed: 37.5 knots

Classification: Germanischer Lloyd IMO
HSC Code B

Main Engines
Model: 16V 4000 M70
Manufacturer: MTU
Number: 4
Output of each engine: 2,320kW at 2,000 rev/min

Generators
Number: 2
Make/type: Cummins
Output: 135kW each

Gearboxes
Make: ZF
Model: BW 755-ID
Number: 4

Waterjets
Make: Kamewa 63 SII
Number: 4

Capacities
Fuel: 24,000 litres
Potable water: 1,500 litres
Sewage: 1,500 litres
Bilge/sludge: 500 litres

Complement
Passengers: 445
Number of cabins: 0

Vehicles
No. of vehicle decks: 1
No. of cars: 20
No. of trucks or trailers: 0

Additional Equipment
Kvaerner motion dampening system;
Eknes Transit Super Nova 1410
passenger seats; Beurteaux Bermuda
110 passenger seats